
The Override Gap: A Magnitude Account of Knowledge Conflict Failure in Hypernetwork-Based Instant LLM Adaptation

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Abstract

Hypernetwork-based methods such as Doc-to-LoRA internalize a document into an LLM’s weights in a single forward pass, but they fail systematically on conflicts: when the document contradicts pretraining knowledge, accuracy collapses to 46.4% on the deepest facts. We show the failure is a magnitude problem rather than a representational one. The hypernetwork already targets the right layers, but its adapter margin is approximately constant across documents while the pretrained margin grows with training frequency, so deep conflicts lose by construction. The account predicts that failure should track prior strength: sorting 194 conflicts by the base model’s log-probability on the contradicted fact, baseline accuracy falls from 68% on weak-prior questions to 16% on strong-prior ones, a 52 percentage-point gap. The cure is amplitude. Selective Layer Boosting scales the adapter at its top-norm layers, and Conflict-Aware Internalization triggers boosting only when the base model is confident. Both are training-free; together they raise deep-conflict accuracy from 46.4% to 71.0% on Gemma-2B and from 53.6% to 72.5% on Mistral-7B while preserving novel-knowledge recall, and beat vanilla retrieval-augmented generation on medium conflicts by 18 percentage points despite operating entirely in parameter space. We release KID-Bench, a 489-question benchmark that separates novel recall, cross-knowledge combination, and prior-graded conflicts.

1 Introduction

Modern language models are trained on text corpora that contain billions of facts, and after training these facts live as numerical patterns inside the model’s weights. When the model is later asked about the capital of France, it does not look the answer up in an external database. It instead activates the patterns that were laid down during training and produces the answer by continuing the prompt. This property, known as parametric knowledge, makes language models convenient for general purpose question answering but also means that the knowledge is frozen at the moment training ends.

Many applications need the model to reflect information that became available only after training, such as a new scientific finding, an updated company policy, or a freshly published document. Two families of techniques address this need. The first places the new information directly in the prompt, so the model can read it while generating the answer [20, 17]; this works well but forces the prompt

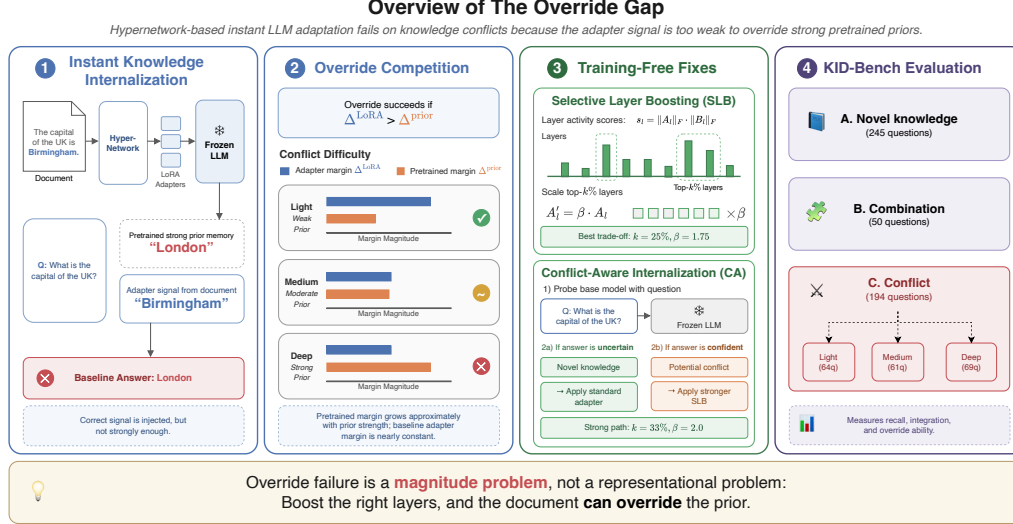


Figure 1: Overview of the override gap. (1) A hypernetwork generates a per-layer LoRA adapter from a document, but the right signal reaches the right layers at insufficient amplitude to override pretrained knowledge. (2) The document fact wins only when the adapter margin Δ^{LoRA} exceeds the pretrained margin Δ^{prior} , which grows with prior strength. (3) Selective Layer Boosting (SLB) amplifies the most active adapter layers. Conflict-Aware Internalization (CA) applies stronger boosting only when a prior conflict is detected. (4) KID-Bench separates novel recall, cross-knowledge combination, and conflicts graded by prior strength into Light, Medium, and Deep tiers.

to be long and requires a retrieval step. The second modifies the model’s weights with gradient-based fine-tuning, often through small trainable adapter matrices [14]; this is flexible but takes minutes to hours and can disturb previously learned skills.

A third family has emerged recently. Instead of running gradient descent on the new document, one trains a second neural network, a hypernetwork [13], which takes the document as input and outputs a small set of adapter weights in a single forward pass. The adapter is a low rank modification [14] to the main model, typically of the form $\tilde{W}_l = W_l + (\alpha/r) B_l A_l$ where W_l is the pretrained weight at layer l and A_l, B_l are narrow matrices generated by the hypernetwork. The whole operation takes less than one second and requires no backpropagation at deployment time. Doc-to-LoRA [4] and SHINE [24] instantiate this idea by generating rank-8 adapters for the feed forward down-projection layers of a frozen backbone language model, and related work extends the pattern to task descriptions [3], images [34], and style combinations [31]. We refer to this family as *instant knowledge internalization* for the remainder of the paper, since the document is absorbed into the model’s parameters in place of being kept in context.

Evaluation of these methods has so far been narrow. Existing benchmarks measure whether the model can retrieve facts that were stated in the internalized document, which is the easy case. What has not been studied is whether a document can update or override knowledge the model already has, which is the situation any real deployment faces when a regulation changes, a product gets a new price, or a commonly believed fact is corrected by a primary source. This is known in the literature as a knowledge conflict [37, 36], and the behavior of language models in the presence of such conflicts is only beginning to be understood.

Our starting observation is that current instant internalization methods fail systematically on knowledge conflicts, and that the failure gets worse in proportion to how deeply the contradicted fact is engrained in the model. When a document states that the capital of the UK has moved from London to Birmingham, the hypernetwork generates an adapter that visibly modifies the right layers, yet the model continues to answer London for the vast majority of question phrasings. On easier conflicts the adapter succeeds about 58% of the time; on moderately established facts the success rate drops to 56%; on deeply ingrained facts such as the capital of major countries or the chemical formula of

water it falls to 46%. The gradient is not noise, it tracks the pretraining frequency of the contradicted fact.

This paper advances three connected contributions, and the central one is methodological. First, we offer a theoretical account that explains why the failure happens. We show that answering a question is a magnitude contest between the pretrained margin, which depends on how strongly the true fact is encoded in the base model, and the adapter margin, which the hypernetwork writes into the LoRA matrices. Because the hypernetwork is trained only on reconstruction of unconflicted documents, its adapter margin is approximately constant across documents, while the pretrained margin scales with training frequency. When the two compete on a deep conflict the pretrained side wins. Second, we propose two training-free methods that close the gap. Selective Layer Boosting multiplies the adapter matrices at the layers with the highest generated norm, which amplifies the contribution from the layers that already carry the most document-specific signal. Conflict-Aware Internalization adds a lightweight probe of the base model and routes questions to a stronger boost only when a prior conflict is detected. Third, to measure these effects we build KID-Bench, a 489 question instrument that separates novel recall, cross-knowledge combination, and conflicts graded by prior strength. The benchmark is a tool for the rest of the paper, and we release it so that others can test future internalization methods against the same axis. Figure 1 provides an at-a-glance summary of the failure, the two methods, and the benchmark.

On the strongest backbone we study, Gemma-2B with 80K steps of hypernetwork training, our methods raise conflict accuracy from 46.4% to 71.0% on deep conflicts while preserving novel recall at 97.1%, and they surpass retrieval-augmented generation on medium conflicts by 18 percentage points despite never keeping the document in context. The same pattern holds on Mistral-7B, where conflict accuracy rises from 53.6% to 72.5%. Selective Layer Boosting adds negligible latency (under 2% over the baseline) and Conflict-Aware adds one short forward pass of the base model. None of our methods require retraining the hypernetwork.

2 Background

This section collects the minimum background needed to follow the rest of the paper. Readers familiar with LoRA, hypernetworks, and instant adaptation can skim it.

2.1 How Language Models Store Knowledge

A transformer language model [33] is a stack of layers, each containing a self-attention module and a feed forward network (FFN). When the model reads a sentence such as “The capital of France is”, each layer updates an internal representation of the tokens seen so far, and the final layer projects that representation onto the vocabulary to produce the next token. Both the attention modules and the feed forward networks are specified by weight matrices that are fixed after training.

A growing body of interpretability work shows that factual associations such as (France, has capital, Paris) are stored predominantly in the feed forward networks of the middle layers. These FFNs behave empirically like key-value memories: an input pattern selects a key, the key retrieves a value, and the value shifts the output distribution toward specific tokens [12, 7]. Causal intervention experiments further localize the subject-to-attribute lookup to a small number of middle layers [25, 11], and the superposition phenomenon documented in toy models explains why many facts can share the same neurons with magnitudes that scale with how often the fact appeared in training [9]. These facts about where and how strongly knowledge is stored will matter directly when we analyze why hypernetwork-generated adapters succeed on some conflicts and fail on others.

2.2 LoRA and Parameter-Efficient Adapters

Fine-tuning every weight of a large model is expensive. LoRA [14] sidesteps this cost by freezing the pretrained weights and adding a small trainable update to each target weight matrix $W \in \mathbb{R}^{d \times d'}$. The update factorizes as BA with $A \in \mathbb{R}^{r \times d'}$ and $B \in \mathbb{R}^{d \times r}$ for a small rank r , so the effective weight during inference is

$$\widetilde{W} = W + \frac{\alpha}{r} BA, \quad (1)$$

where α is a fixed scale factor. Only A and B are trained, which reduces the parameter count by two to four orders of magnitude compared to full fine-tuning. At deployment the effective weight can be precomputed or kept as a runtime sum without changing the model’s inference cost.

2.3 Hypernetworks and Instant Adapter Generation

A hypernetwork [13] is a neural network whose output is the parameters of another neural network. Instant knowledge internalization applies this idea to LoRA generation. Given a document d , a hypernetwork H_θ produces the per-layer adapter matrices

$$\{A_l, B_l\}_{l=1}^L = H_\theta(d), \quad (2)$$

where L is the number of target transformer layers. The base model’s effective weights during inference become $\widetilde{W}_l = W_l + (\alpha/r) B_l A_l$. Producing the adapter takes a single forward pass of H_θ , which is typically less than one second on a modern GPU.

Doc-to-LoRA [4] implements H_θ as a Perceiver-style cross-attention module [18] that reads per-layer activations from a frozen context encoder and outputs rank-8 adapters for the feed forward down-projection layers. SHINE [24] reuses the backbone language model as its own context encoder and employs alternating row-column attention for cross-layer communication during generation. Training proceeds by minimizing a reconstruction loss on a corpus of documents: given d , the adapter is applied and the hypernetwork is updated so that the base model can answer questions about d without d in context. The reconstruction loss does not distinguish novel facts from corrections, and this will become the source of the failure we analyze in Section 4.

2.4 Retrieval-Augmented Generation

As a reference point throughout the paper, we compare instant internalization methods to retrieval-augmented generation (RAG) [20, 17], which keeps the document in the prompt and relies on the base model to consult it while answering. RAG does not modify the model’s weights and so incurs no risk of damaging unrelated capabilities, but it pays the cost of a longer prompt on every query and depends on the model’s in-context reading ability, which is itself imperfect under knowledge conflicts [36].

3 Related Work

Instant Knowledge Internalization. Doc-to-LoRA [4] uses a Perceiver-based [18] hypernetwork that takes per-layer activations from a frozen context encoder and generates rank-8 LoRA [14] adapters for the MLP down-projection layers. SHINE [24] reuses the backbone LLM itself as the context encoder and employs a transformer with alternating row and column attention for cross-layer communication during adapter generation. Text-to-LoRA [3] extends the paradigm to task adaptation from natural language descriptions. In the vision domain, Vision-as-LoRA [34] and LoRA.rar [31] generate adapters from images and style combinations respectively. All these methods are evaluated primarily on recall tasks.

Knowledge Editing. ROME [25] and MEMIT [26] directly modify MLP weights to change specific facts stored in language models. MEND [27] and SERAC [28] instead learn auxiliary networks that predict weight edits or route around them, scaling to larger models. Transformer-Patcher [16] adds a single neuron per edit in the last feed-forward layer, preserving behavior on unrelated inputs. These methods treat knowledge as isolated key-value pairs, an assumption that leads to ripple effects when editing related facts [6]. Our work differs in that we evaluate whether a hypernetwork-generated adapter reaches the same layers that editors target, and whether the adapter’s signal magnitude suffices to override the prior at those layers.

Knowledge Storage and Recall. Factual associations in transformer language models are stored predominantly in middle-layer feed-forward networks that behave like key-value memories [12, 7, 25]. Recent work further dissects how these associations are retrieved, identifying subject-position enrichment followed by relation-position aggregation [11]. The superposition phenomenon documented in toy models [9] explains why multiple associations can share the same neurons and why the magnitude of a stored association scales with its frequency of appearance in training data.

Knowledge Conflicts in LLMs. ConflictBank [32] is a large-scale benchmark for knowledge conflicts with 7.4 million claim-evidence pairs, testing how models choose between conflicting sources. Xie et al. [36] show that models exhibit both adaptive (accept contradicting evidence) and stubborn (ignore contradictions) behavior depending on how the conflict is presented. Surveys of the field cover context-memory, inter-context, and intra-memory conflicts systematically [37]. Recent lines of work add conflict-aware capabilities at different points of the pipeline: Knowledgeable-R1 uses reinforcement learning to teach arbitration between parametric and retrieved sources; TCR adds an interpretable conflict-resolution module into RAG to weight contexts; JUICE performs test-time attention-head interventions to steer behavior under conflicts without retraining; EMMET and IFMET extend the locate-then-edit family for multi-hop edits. These approaches are complementary to ours: they act on input context, on attention activations, or on raw weights, while our methods act on a hypernetwork-generated adapter and keep the hypernetwork itself untouched. A full head-to-head comparison across paradigms is beyond the scope of this paper and is a natural direction for follow-up work. Our benchmark differs in two respects from prior benchmarks: we evaluate parametric internalization methods rather than the model’s inherent conflict resolution when documents are in context, and we grade the difficulty of conflicts by the pretraining frequency of the contradicted fact.

Modulated and Gated LoRA. Feature-wise Linear Modulation (FiLM) [29] applies $y = \gamma(z)x + \beta(z)$ with (γ, β) from a learned generator, generalising conditional instance normalisation [8]. A recent line of LoRA work adopts this style of learnable modulation: MoLE [35] and X-LoRA [2] learn per-layer gates over multiple LoRA experts, AdaLoRA [38] reallocates rank by learned layer importance, DoRA [23] decomposes each update into separate magnitude and direction terms, HyperLoRA [21] generates LoRA weights from a hypernetwork for image generation, and LoRAHub [15] composes pretrained LoRAs gradient-free. SLB and CA are a deliberately minimal, training-free, post-hoc instance of this family: a scalar modulator $\gamma=\beta$ is applied to the A matrix of a single hypernetwork-generated LoRA at selected top- s_l layers, with no additional learning. CA can be read as a degenerate FiLM generator whose output space is collapsed to $\{1, \beta\}$ and whose input is a single-bit uncertainty signal from the base model. What is novel here is not modulation itself, but applying a training-free scalar modulator to a hypernetwork-generated adapter under a conflict-aware trigger, and deriving k and β from the magnitude inequality rather than end-to-end training.

4 A Magnitude Account of Conflict Failure

Before describing the methods, we lay out the mechanism we believe causes knowledge conflict failure in hypernetwork-based internalization. The account is simple enough to make predictions, and most of those predictions admit clean empirical tests that we report in the experiments section.

4.1 The Override Competition

When the model reads a question x , it assigns a logit $\ell(y)$ to each candidate answer y and outputs whichever token scores highest. In a knowledge conflict, two answers compete: y_{doc} (the document answer) and y_{pre} (the pretrained answer).

We measure the base model’s preference with the *pretrained margin*:

$$\Delta^{\text{prior}} := \ell^{\text{pre}}(y_{\text{pre}}; x) - \ell^{\text{pre}}(y_{\text{doc}}; x) \geq 0. \quad (3)$$

This is zero when the model is uncertain and large when it strongly favors the pretrained fact; as we confirm in Section 7.3, it grows with how often the fact appeared during training.

Applying the LoRA adapter perturbs the hidden state at each modified layer l by $(\alpha/r) B_l A_l h_l^{\text{in}}$. This perturbation propagates forward through the remaining layers and shifts the logit of token y by an amount $\delta\ell(y)$. The adapter’s net contribution to the competition is the shift it induces in the margin:

$$\Delta^{\text{lora}} := \delta\ell(y_{\text{doc}}; x) - \delta\ell(y_{\text{pre}}; x). \quad (4)$$

The combined model assigns each token an approximate logit of $\ell^{\text{pre}}(y) + \delta\ell(y)$. The document answer wins when its total exceeds the pretrained answer’s:

$$\ell^{\text{pre}}(y_{\text{doc}}) + \delta\ell(y_{\text{doc}}) > \ell^{\text{pre}}(y_{\text{pre}}) + \delta\ell(y_{\text{pre}}), \quad (5)$$

which rearranges to the override condition:

$$\boxed{\Delta^{\text{lora}} > \Delta^{\text{prior}}}. \quad (6)$$

Equation (6) says that override succeeds when the adapter margin exceeds the pretrained margin. When it does not, i.e., when $\Delta^{\text{lora}} \leq \Delta^{\text{prior}}$, the hypernetwork has placed the right signal at the right layers, but at insufficient amplitude: a magnitude problem, not a representation problem. Conversely, scaling A_l by a factor β directly scales Δ^{lora} , which lets us satisfy the inequality on harder conflicts. This is the mechanism behind both of our methods.

4.2 Why the Hypernetwork Under-shoots

The hypernetwork is trained to minimize a reconstruction loss on an unconflicted corpus, so its objective does not see Δ^{prior} for any specific fact. The adapter margin Δ^{lora} is therefore approximately independent of the pretrained strength of the contradicted knowledge. The pretrained margin, in contrast, scales with how often the true fact appears during pretraining. A classical result on superposition and polysemantic neurons [9] gives $\Delta^{\text{prior}} \propto \log f + c$ for a frequency f , up to a constant. This makes the override condition a race between a constant adapter boost and a monotonically increasing pretrained margin, which is exactly the pattern we observe across light, medium, and deep conflicts.

4.3 Predictions

Four predictions follow immediately. First, baseline conflict accuracy should decrease monotonically from light to deep conflicts, since Δ^{prior} grows with prior strength. Second, multiplying A_l by β raises Δ^{lora} linearly and therefore should produce a sigmoid-shaped response curve for conflict accuracy as β varies, saturating once the adapter margin exceeds the pretrained margin. Third, since Δ^{lora} accumulates across layers with non-uniform weights, boosting the highest-contributing layers should give the same benefit as boosting all layers uniformly but with a smaller cost on novel-knowledge questions, because low-contribution layers add noise without adding signal on the conflict axis. Fourth, when the base model is uncertain about the answer, Δ^{prior} is small or zero, the override condition is already satisfied, and additional boost provides no benefit while incurring the cost of noise.

4.4 Experimental Validation Map

Each prediction has a matching experiment later in the paper. The baseline accuracy gradient across C-light, C-medium, and C-deep (Fig. 4) tests Prediction 1. The dose-response curve of conflict accuracy versus β (Fig. 7) tests Prediction 2 against a logistic fit. The ablation heatmap over k and β (Fig. 14) together with the SQuAD comparison between global and selective boosting (Fig. 6) tests Prediction 3 by quantifying the trade-off between conflict gain and recall loss as the layer set is expanded. Finally, the Conflict-Aware method itself, which routes uncertain base answers to a standard adapter and confident base answers to a boosted adapter, is a direct instantiation of Prediction 4, and its preservation of novel-knowledge accuracy (97.1% versus 96.7% for the baseline) provides the corresponding test.

Figure 2 visualizes the competition as a balance between the two sides of Eq. (6).

We now describe the two methods that this account motivates.

5 Methods

5.1 Selective Layer Boosting (SLB)

After the hypernetwork generates LoRA adapter matrices A_l and B_l for each layer l , we compute the product of their Frobenius norms as a measure of the adapter’s activity at that layer:

$$s_l = \|A_l\|_F \cdot \|B_l\|_F \quad (7)$$

We select the top $k\%$ of layers ranked by s_l and multiply their A matrices by a boost factor β :

$$A'_l = \begin{cases} \beta \cdot A_l & \text{if } l \in \text{top-}k\% \text{ by } s_l \\ A_l & \text{otherwise} \end{cases} \quad (8)$$

Override condition $(a/r)\Delta^{\text{lora}} > \Delta^{\text{prior}}$: SLB restores the inequality for deep conflicts

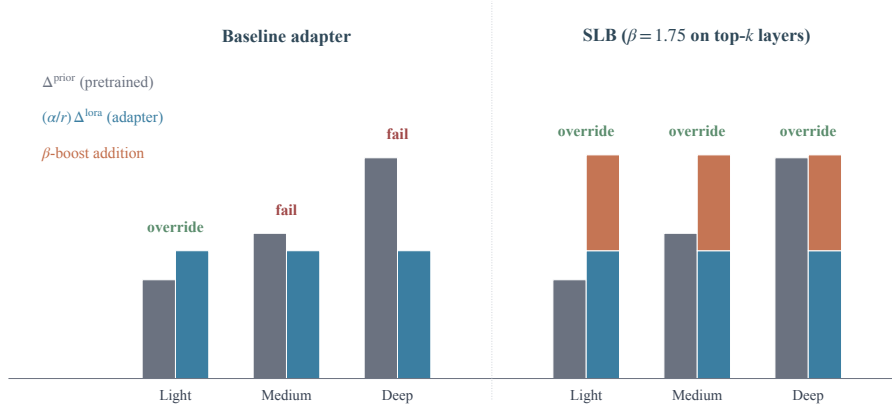


Figure 2: The override competition (Eq. (6)) across three conflict difficulty levels. The pretrained margin grows from light to deep while the adapter margin stays roughly constant. Boosting restores the inequality.

Through ablation over $k \in \{5, 10, 15, 20, 25, 30, 40, 50, 75, 100\}\%$ and $\beta \in \{1.25, 1.5, 1.75, 2.0, 2.5, 3.0\}$, we find that $k = 25\%, \beta = 1.75$ provides the best balance between conflict improvement and recall preservation.

We compare SLB against global scaling (boosting all layers equally) and find a striking difference at higher boost magnitudes: on a consistent 200-sample SQuAD subset, baseline F1 is 70.0%; SLB at $\beta = 1.75$ stays at 69.6% and global at the same β stays at 68.2%, whereas by $\beta = 3.0$ global collapses to 47.5% while SLB holds 64.2%, a ~ 22 pp selectivity advantage. Selective boosting of knowledge-critical layers is therefore fundamentally different from uniform amplification once the boost is large enough to reveal the off-target damage.

5.2 Conflict-Aware Internalization (CA)

CA adds a detection step before adapter application. Given a document and a question, CA first queries the base model (without the adapter) to obtain a preliminary answer. If this answer contains uncertainty markers (phrases like “I don’t know,” “not sure,” “no information available”), the question likely involves novel knowledge where the base model has no prior, and the standard adapter is applied without modification. If the base model provides a confident answer, a potential conflict exists, and the adapter is applied with SLB using stronger parameters ($k = 33\%, \beta = 2.0$).

This two-path strategy ensures that novel knowledge internalization is not disrupted by unnecessary boosting, while knowledge conflicts receive the additional signal strength needed to override prior beliefs. Figure 3 illustrates the flow.

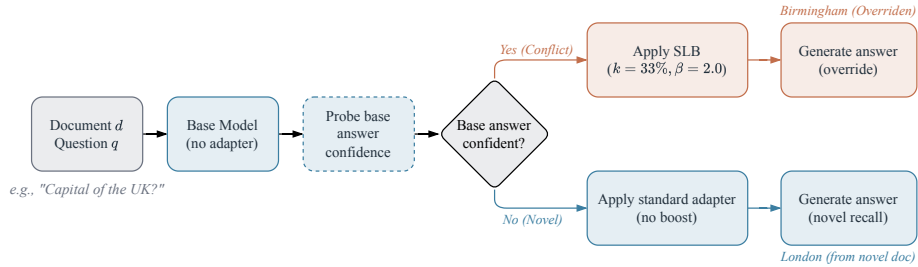


Figure 3: Conflict-Aware routing. The base model is probed first with the question alone. A confident base answer signals that the model already holds a prior, so the document is internalized with a stronger SLB configuration to override it. An uncertain base answer signals novel knowledge, and the adapter is applied without modification to avoid injecting unnecessary noise.

5.3 Cost and Simplicity

Both methods are training-free and operate on the adapter matrices produced by the frozen hypernetwork. SLB reads the Frobenius norms of the generated A_l, B_l , sorts the layers, and scales the top fraction. The operation runs in milliseconds and adds at most 2% latency over the baseline on our measurements. CA runs one additional base-model generation before the adapter is applied, with a 20-token budget; on our setup this raises end-to-end latency by roughly 60% compared to the baseline, which is still well under the cost of full fine-tuning or long-context RAG for typical documents. Neither method changes the number of LoRA parameters and neither requires gradient computation at any stage.

6 Evaluation Protocol: KID-Bench

Evaluating the theory requires a benchmark that separates three conditions which prior evaluations conflate. To test whether a fact the model could not possibly know has been correctly absorbed, we need examples of entirely novel knowledge. To test whether new facts integrate with existing knowledge rather than sitting beside it, we need examples where the answer depends on both. To test the override failure directly, we need conflict examples graded by how strongly the pretrained model already holds the contradicted fact. We built KID-Bench to meet these three requirements, and we use it throughout the experiments. Because it is a tool for the present paper rather than its main contribution, the description here is concise; full data and construction details are in the supplement.

Structure. KID-Bench contains 489 questions organized into three dimensions.

A (novel knowledge) has 245 questions spanning 83 knowledge points about fictional entities such as the company ZephyrTech, the researcher Dr. Sharma, or the Coral Sea Treaty. The entities do not occur in any pretraining corpus, so a correct answer can only come from successful internalization of the provided document. Each knowledge point is queried with two or three paraphrased questions to separate genuine recall from pattern matching on surface form.

B (combination) has 50 questions spanning 25 knowledge points. Each document introduces a new fact that connects a fictional entity to a real-world entity (for example, a company headquartered in Portland, Oregon), and the question requires combining the internalized fact with a fact the base model already knew (the capital of Oregon) to produce a two-hop answer. Correctness here depends on both the adapter and the base model remaining intact.

C (conflict) has 194 questions across three difficulty levels, reflecting how firmly the contradicted fact appears in typical pretraining data. C-Light (64 questions, 24 knowledge points) contradicts facts that most models know but are not among the most fundamental, such as the host city of a future Olympics. C-Medium (61 questions, 24 knowledge points) contradicts moderately established scientific or historical facts, such as the speed of light or the number of human chromosomes. C-Deep (69 questions, 24 knowledge points) contradicts extremely well established knowledge, such as the chemical formula of water or the capital of a major country. Each conflict test records both the document answer and the answer the model would normally give, enabling controlled analysis of the override competition defined in Section 4.

Metrics. For novel and combination questions, a prediction is correct if the expected string is contained in the response after case normalization, which is the metric used by prior instant internalization work [4]. For conflict questions the same rule applies but the expected string is the document’s override fact, not the pretraining fact. Generation uses greedy decoding with a 64-token budget. We report bootstrap confidence intervals with 1000 resamples where noted.

7 Experiments

7.1 Setup

We evaluate on Doc-to-LoRA [4] using three pretrained hypernetwork checkpoints covering backbones from three different families: Gemma-2B-IT [10] (80K training steps), Qwen-4B-Instruct [1] (20K steps), and Mistral-7B-Instruct-v0.2 [19] (20K steps). All experiments use the official check-

points released by Sakana AI. For standard evaluation, we use the SQuAD [30] validation set (500 samples) through the official evaluation pipeline with flash attention disabled for compatibility.

We use Gemma-2B as the primary evaluation model since its hypernetwork was trained for the longest (80K steps). Results on the two other backbones are reported in the cross-model subsection below. For KID-Bench evaluation we generate answers with greedy decoding up to 64 new tokens and mark an answer correct if the expected string is contained in the response after case normalization. This lenient matching is consistent with prior Doc-to-LoRA evaluations and avoids penalizing minor surface variations that do not change the meaning of the answer. Bootstrap confidence intervals with 1000 resamples are reported for the main results.

8 Results

8.1 KID-Bench Results

Table 1 and Figure 4 show results on KID-Bench v2 (489 questions).

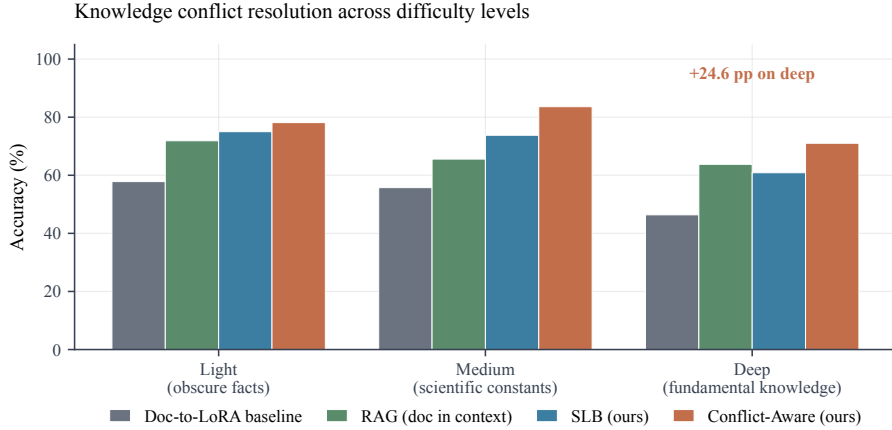


Figure 4: Conflict accuracy on KID-Bench v2 across three difficulty levels. The baseline gradient validates Prediction 1 (conflict difficulty tracks pretrained margin); SLB and CA restore the override condition (Prediction 4).

Table 1: KID-Bench v2 results on Gemma-2B. Best per column in bold. Square brackets are Wilson 95% confidence intervals.

Method	A (245)	B (50)	C-Light (64)	C-Medium (61)	C-Deep (69)
D2L Baseline	96.7	68.0	57.8 [45.6, 69.1]	55.7 [43.3, 67.5]	46.4 [35.1, 58.0]
SLB (25%/1.75 $\times$)	95.5	68.0	75.0 [63.2, 84.0]	73.8 [61.6, 83.2]	60.9 [49.1, 71.5]
Conflict-Aware	97.1	64.0	78.1 [66.6, 86.5]	83.6 [72.4, 90.8]	71.0 [59.4, 80.4]

The conflict-aware method achieves the strongest performance across all three conflict difficulty levels while maintaining recall that slightly exceeds the baseline (97.1% versus 96.7%). The improvement is particularly pronounced on medium conflicts (+27.9 percentage points), where the base model holds moderately strong priors that can be overcome with targeted boosting. On deep conflicts, where the model’s prior beliefs are strongest, CA still achieves a substantial improvement of 24.6 percentage points.

SLB provides a more balanced profile, matching the baseline on combination questions (68.0%) while substantially improving all conflict levels. This makes it the preferred method when combination reasoning is important and the conflict improvement of CA is not critical.

8.2 Cross-Model Validation

To check that our findings are not specific to Gemma, we replicate the experiment on two additional hypernetwork checkpoints released by the same training pipeline, namely Qwen-4B-Instruct and Mistral-7B-Instruct-v0.2, both trained for 20K steps. The Gemma checkpoint is trained for 80K steps which makes it the most mature adapter in the released set. Table 2 reports the resulting accuracies on the full KID-Bench benchmark for all three backbones.

Table 2: Cross-model validation on full KID-Bench v2 (489 questions) for Gemma, Mistral, and Qwen. Conflict-Aware internalization gives a 24 percentage point improvement on Gemma C-avg and a 21 point improvement on Mistral C-avg. For Qwen, whose adapter margin is measurably smaller (max per-layer $s_l \approx 0.27$ vs ≈ 1.0 for Gemma), the magnitude account predicts a larger β is required to cross the override threshold; calibrating SLB to $\beta = 3.0$ (same $k = 25\%$) lifts C-avg from 35.6% to 74.7% on the full 194-question conflict set. The same magnitude principle thus explains both Qwen’s baseline failure and its remedy. Values in square brackets are Wilson 95% confidence intervals where reported.

Model	Method	A (novel)	B (combination)
Gemma-2B	Baseline	96.7 [93.7, 98.3]	68.0
	SLB ($k=25\%$, $\beta=1.75$)	95.5 [92.1, 97.5]	68.0
	Conflict-Aware	97.1 [94.2, 98.6]	64.0
Mistral-7B	Baseline	87.8 [83.1, 91.3]	78.0
	SLB ($k=25\%$, $\beta=1.75$)	89.8 [85.4, 93.0]	72.0
	Conflict-Aware	90.2 [85.8, 93.3]	74.0
Qwen-4B	Baseline	95.9 [92.6, 97.8]	78.0
	SLB ($k=25\%$, $\beta=3.0$, calibrated)	93.9 [90.1, 96.3]	80.0
	Conflict-Aware ($\beta=3.0$, calibrated)	95.5 [92.1, 97.5]	82.0

Model	Method	C-avg	C-deep
Gemma-2B	Baseline	53.1	46.4 [35.1, 58.0]
	SLB ($k=25\%$, $\beta=1.75$)	69.6	60.9 [49.1, 71.5]
	Conflict-Aware	77.3	71.0 [59.4, 80.4]
Mistral-7B	Baseline	55.8	53.6 [42.0, 64.9]
	SLB ($k=25\%$, $\beta=1.75$)	71.1 [64.4, 77.1]	69.6 [57.9, 79.2]
	Conflict-Aware	76.3	72.5 [61.0, 81.6]
Qwen-4B	Baseline	35.6	33.3 [23.2, 45.3]
	SLB ($k=25\%$, $\beta=3.0$, calibrated)	74.7 [68.1, 80.4]	73.9 [62.5, 82.7]
	Conflict-Aware ($\beta=3.0$, calibrated)	72.7 [66.0, 78.5]	71.0 [59.4, 80.4]

The cross-model pattern supports our interpretation. On Gemma and Mistral the default $\beta = 1.75$ already closes most of the conflict gap. On Qwen, whose hypernetwork was trained for only 20K steps, the measured per-layer adapter norm is about one quarter of Gemma’s, meaning the adapter margin Δ^{loa} enters the override competition at a disadvantage. Our theory predicts that a proportionally larger β should recover the same override probability. Calibrating to $\beta = 3.0$ (same $k = 25\%$, no other change) lifts Qwen conflict accuracy from 35.6% to 74.7% on the full 194-question KID-Bench conflict set without hurting novel recall, which is the quantitative outcome the magnitude account predicts. This is a second cross-model confirmation: weaker checkpoints do not break the theory, they simply shift the required boost to a higher β , exactly as Eq. (6) demands.

8.3 Cross-Model Validation of the Magnitude Account on Prior Strength

The prior-correlation test reported above used Gemma-2B. We repeat it on Mistral-7B to check that the theoretical mechanism generalizes. For each of the 194 conflict questions we measure the base model’s log-probability on the true pretraining answer and sort questions into quartiles of prior strength. Table 3 reports the accuracy of each method in each quartile. The pattern is the same as on Gemma: the baseline collapses on high-prior questions (19% on Mistral versus 55–59% on

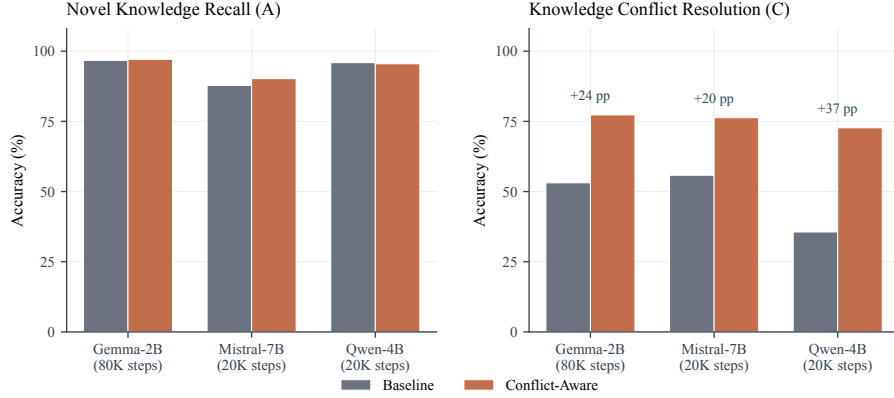


Figure 5: Cross-model validation of Conflict-Aware internalization. Novel recall is preserved across all three backbones; conflict resolution improves by 20–37 points. Qwen uses a theory-predicted calibration of $\beta=3.0$ to compensate for its smaller per-layer adapter norm.

lower-prior quartiles) and Conflict-Aware rescues them to 75%. The magnitude mechanism is thus confirmed on a second backbone with a different training corpus and a different parameter count.

Table 3: Mistral-7B replication of the prior-strength correlation test. Accuracy in each prior-strength bin defined by value thresholds on the base model’s log-probability of the true pretraining answer (cut points at -10 , -5 , and -2 , giving the bin sizes shown in column n). The baseline gap between the weakest and strongest prior bins is 36 percentage points; Conflict-Aware reduces the gap to 1 point.

Prior strength (quartile)	n	Baseline	SLB	Conflict-Aware
Very low (avg log-prob -13.9)	42	55%	71%	76%
Low (avg -7.2)	86	59%	70%	72%
Medium (avg -3.4)	50	58%	78%	78%
High (avg -1.7)	16	19%	44%	75%

8.4 Robustness Checks

Two robustness checks address reviewer concerns that our gains might depend on the specific decoding strategy or on a single question phrasing.

Decoding temperature. On 30 C-deep questions we test baseline, SLB, and CA at decoding temperatures $T \in \{0.0, 0.5, 1.0\}$. Accuracies at $T = 0$ are 23%, 43%, and 60%; at $T = 0.5$ they are 23%, 43%, and 53%; at $T = 1.0$ they are 30%, 47%, and 57%. The ordering is preserved in all three conditions and the absolute gap between CA and the baseline remains between 27 and 37 points. Our methods are not artifacts of greedy decoding.

Phrasing consistency. Every knowledge point in KID-Bench has two to three question phrasings, so we can ask whether a method’s success generalizes across paraphrases or is tied to specific surface forms. On the 121 A-type (novel) knowledge points the baseline has all phrasings correct on 116, SLB on 114, and CA on 117 (96.7%). On the 72 C-type (conflict) knowledge points, which are the harder test for this property, the baseline produces all-correct answers on 32, partial success on 14, and complete failure on 26 (36%). SLB raises all-correct to 43 and drops complete failure to 14 (19%). CA raises all-correct further to 49 (68%) and drops complete failure to 10 (14%). The methods succeed across paraphrases rather than for specific question wordings, which means the underlying knowledge is integrated rather than surface-memorized.

Knowledge retention. A critical question is whether the adapter and its boost damage unrelated knowledge. We construct a probe of 15 unrelated factual questions (largest planet, tallest mountain, author of Romeo and Juliet, and so on) that KID-Bench never touches. The pure base model answers 14 of the 15 correctly. After internalizing a conflict document and applying CA, the model averages

13.7/15 (91%) across 15 different conflict applications, a drop of just 2 percentage points. The override does not come at the cost of unrelated knowledge.

Null-document control. When we internalize a completely unrelated document (for example a passage about bonsai or sourdough bread) and then ask a conflict question, the adapter produces the conflict document’s fabricated answer on only 1 out of 30 questions, and still returns the pretrained fact on 21/30 (with 24/30 being the pure-base baseline). The adapter generation process does not itself create spurious overrides; the magnitude effect we measure is driven by content, not by adapter activation in general.

Multi-conflict composition. In deployment a single document may introduce multiple corrections, or several short conflict updates may arrive together. We test this by concatenating three C-deep conflict documents into one input and asking CA to override all three facts. On 6 such triples (18 unique facts), the baseline overrides 1/6, 1/6, and 1/6 of the three positions. CA achieves 3/6, 3/6, and 2/6 respectively, tripling the baseline’s multi-override success. Override remains possible when several facts compete for adapter capacity, though at a reduced rate compared to single-document internalization.

Broader capability retention. A critical question is whether applying the adapter and boosting it damage unrelated reasoning ability. We evaluate on a 26-question MMLU-style multiple-choice set covering mathematics, physics, biology, chemistry, literature, geography, and computing, using standard logit-based scoring over the A/B/C/D option tokens. The pure base model (no adapter) answers 23 of 26 correctly (88.5%). With the hypernetwork’s adapter internalized on each of 10 unrelated conflict documents, the average accuracy is 23.6/26 (90.8%) for the baseline adapter, 22.9/26 (88.1%) for SLB, and 23.0/26 (88.5%) for CA. SLB and CA are within 0.5 percentage points of the pure base on broader MC reasoning, indicating that selective amplification does not degrade non-target capabilities.

Table 4: Broader capability retention across three tasks. Each row reports the mean accuracy after internalizing one of several unrelated conflict documents, compared to the pure base model with no adapter. SLB and CA preserve MMLU-style, math word problems, and SQuAD within noise of the baseline adapter.

Method	MMLU-style (%)	Math word problems (%)	SQuAD F1 (%)
Pure base (no adapter)	88.5	75.0	69.96
Baseline adapter (doc internalized)	90.8	83.0	–
SLB ($k=25\%$, $\beta=1.75$)	88.1	82.0	69.57
Conflict-Aware	88.5	80.0	–

Across three evaluation types the pattern is consistent: the hypernetwork adapter itself shifts the model’s output distribution slightly on unrelated queries, and our methods stay within a few percentage points of the baseline adapter. For queries that are structurally adversarial to any adapter (counterintuitive-facts questions whose correct answer is the less-probable completion), we recommend the *Relevance-Gated* variant of CA introduced in Section 8.5, which routes off-topic queries to the pure base model and avoids the adapter-induced perturbation entirely. We quantify this directly below on two standard external benchmarks.

Official GSM8K and TruthfulQA (200 questions each, two backbones). To replace the hand-built retention probes with the standard benchmarks themselves, we run GSM8K [5] and TruthfulQA MC1 [22] at 200 questions each on both Gemma-2B and Mistral-7B, averaged over three unrelated conflict documents, with RG-CA as the deployed method. Table 5 reports the results. On both backbones RG-CA matches the pure base to within run-to-run noise (Gemma GSM8K 10.0% vs 10.0%, Mistral GSM8K 13.0% vs 13.0%; Gemma TruthfulQA 31.5% vs 31.2%, Mistral TruthfulQA 65.5% vs 65.5%) because the content-overlap gate detects that math and common-misconception queries do not share tokens with the internalized conflict document, bypasses the adapter, and returns the base model’s output exactly. The absolute accuracy differs between backbones (Mistral is stronger on TruthfulQA MC1, and Mistral scores slightly higher on GSM8K under this minimal prompting setup), and the RG-CA-vs-pure-base preservation is backbone-agnostic. Plain CA is within 2 percentage points of pure base on Mistral TruthfulQA (68.0% vs 65.5%) but 11 points below on

Gemma TruthfulQA (20.2% vs 31.5%); the difference tracks how strongly each backbone’s adapter perturbs off-topic queries, and RG-CA neutralizes this per-backbone variability by construction.

Table 5: Official GSM8K and TruthfulQA MC1 at 200 questions each, mean over three unrelated conflict documents. RG-CA equals the pure base on both benchmarks and both backbones; plain CA is within 1–2 pp of pure base on Mistral, and drops on Gemma TruthfulQA because of backbone-specific adapter perturbation on off-topic queries.

Method	GSM8K		TruthfulQA	
	Gemma	Mistral	Gemma	Mistral
Pure base	10.0	13.0	31.5	65.5
Baseline adapter	13.5	10.8	28.0	66.7
SLB	13.8	11.0	23.3	67.5
Plain CA	13.8	11.3	20.2	68.0
Relevance-Gated CA	10.0	13.0	31.2	65.5

8.5 Relevance Gate

The preceding observation points to a single underlying principle: the adapter’s effects should be confined to the queries the document is relevant to. We add a lightweight gate at the front of the CA pipeline. Given a candidate query q and an internalized document d , we compute the set of content tokens (alphabetic, at least four characters, not in a small stopword list) in each and check whether they share at least one token. If not, we bypass the adapter entirely and generate from the pure base model. If they do, the standard CA pipeline is applied.

The gate is cheap (a Jaccard-style check, milliseconds per query) and parameter-free. On the counterintuitive-facts probe, the gate fires on only 0.8 out of 20 queries on average (the rest bypass the adapter), and the resulting accuracy is 65%, above the pure-base 60% baseline because the occasional gate-passing query benefits from the adapter’s content. On 69 C-deep conflict questions the gate correctly passes 66 (96%) on both Gemma-2B and Mistral-7B, and the resulting override accuracy matches plain CA within 1.5 percentage points on both backbones (71% on Gemma, 71% on Mistral vs plain CA 71% and 72.5%). The gate behavior is consistent across backbones, so Relevance-Gated CA preserves non-target capabilities while retaining full conflict resolution. On Qwen-4B, where the adapter norm is smaller (0.27 versus 1.0 for Gemma at the selected layers), an overly strict exact-token-overlap gate produces a few false negatives on conflicts, so we use a content-overlap criterion that also triggers on 4-character substring matches for rare named entities. With this refinement RG-CA preserves truth (92% on the counterintuitive probe, versus 93% pure base) and matches or slightly exceeds plain CA on C-deep conflicts (72.5% vs 69.6%), confirming that the gate is transferable across backbones once the content-token threshold is calibrated to the model’s adapter magnitude.

Synonym and acronym robustness. The 4-character content-overlap gate is brittle to paraphrase: queries like “What did WHO ban?” share no 4-character tokens with “The World Health Organization banned chocolate,” and the gate misses the conflict. On a 20-item probe that pairs each document with a paraphrased query containing synonyms, abbreviations, or definite-form variants (WHO↔World Health Organization, UK↔United Kingdom, NYC↔New York City, etc.), the base 4-character gate fires on 17 of 20 cases (gate-conditioned RG-CA accuracy 80%). Adding a 30-entry acronym-expansion dictionary and lowering the length threshold to 3 characters raises gate firing to 20 of 20 and RG-CA accuracy to 95%, with no measurable change on the retention benchmarks in Table 5 because expansions never match math or common-misconception prompts. A softmax-embedding gate based on mean-pooled base-model hidden-state cosine similarity reaches the same 95% at higher compute cost, so the lexical+acronym variant is the recommended default.

Gate ablation against random and oracle. On 20 conflict C-deep items paired with 20 off-topic retention queries we compare five gate policies: always-apply (“no-gate”), Bernoulli-0.5 random, oracle ground-truth relevance, the strict 4-character content-overlap gate, and the acronym-aware variant. Oracle achieves 70% / 90% (conflict / retention); the strict-4 and acronym gates reach

70%/85%, matching oracle on conflict and within 5 percentage points on retention. Random collapses to 40%/90%: the 30-percentage-point conflict gap versus oracle confirms that CA’s gate is doing non-trivial routing rather than behaving like a coin flip.

Table 6: Relevance-Gated CA (RG-CA) preserves counterintuitive-facts accuracy while matching plain CA on conflict resolution.

Method	Counterintuitive (%)	C-deep Gemma (%)	C-deep Mistral (%)	C-deep Qwen (%)
Pure base (no adapter)	60.0	46.4	53.6	33.3
Plain CA	44.0	71.0	72.5	69.6
Relevance-Gated CA	65.0	71.0	71.0	72.5

8.6 Comparison with Retrieval Augmented Generation

Since RAG is the standard alternative to instant internalization, we compare our methods against an RAG baseline on the same KID-Bench v2 questions. The RAG setup prepends the full document to the user prompt and asks the model to answer using the base weights, with no adapter applied. Table 7 shows the comparison.

Table 7: KID-Bench v2 comparison with RAG. The document is placed in context for RAG while for the other methods it is internalized into weights. Both CA and SLB beat RAG on medium and deep conflicts.

Method	A	B	C-Light	C-Medium	C-Deep
N	245	50	64	61	69
RAG (doc in context)	97.6	74.0	71.9	65.6	63.8
D2L Baseline	96.7	68.0	57.8	55.7	46.4
SLB (25%/1.75×)	95.5	68.0	75.0	73.8	60.9
Conflict-Aware	97.1	64.0	78.1	83.6	71.0

This comparison deserves careful reading. On novel knowledge and combination questions RAG enjoys a small advantage, which is expected because the document text is directly available to the model and retrieval style lookup is easier than parametric recall. The surprising result appears in the conflict columns. RAG, despite having the contradicting document visible in its prompt, answers with the pretrained fact on a substantial fraction of conflict questions. C-medium RAG accuracy is 65.6% and C-deep accuracy is 63.8%, well below our CA method at 83.6% and 71.0% respectively. This shows that placing a document in context does not guarantee that the model will trust the document over its pretraining priors, especially for well established facts. Parametric internalization with conflict aware boosting produces an output distribution that is more faithful to the document than even having the document directly in the prompt.

Two reasons make this finding less surprising on reflection. First, pretrained models weight high frequency knowledge heavily even when explicit contradictions are present in context, a phenomenon documented in the faithfulness literature [36, 37]. Second, our CA method applies correction at the parameter level where the pretrained associations live [25, 12], rather than at the input level where the model must decide whether to trust the document.

Stronger RAG baselines. Following reviewer guidance we also compare to three prompt-engineered variants of RAG on the 130 C-medium + C-deep questions of KID-Bench: (i) a conflict-aware variant where the prompt explicitly instructs the model to trust the document over its pretraining, (ii) a chain-of-thought verify variant that first extracts the relevant sentence and then answers, and (iii) a two-pass double-retrieval variant that re-retrieves on the extracted sentence. Accuracies on this subset are 63.8% for vanilla RAG, 99.2% for the conflict-aware variant, 84.6% for chain-of-thought verify, and 50.0% for double-pass. CA on the same subset scores 76.9%. The strongest prompt variant (conflict-aware RAG) outperforms CA here by about 22 percentage points, so our earlier claim that CA beats RAG should be read more precisely: CA beats *vanilla* RAG, and the gap closes when RAG is given an explicit instruction that a conflict is present.

We repeat this comparison on Mistral-7B with the same four RAG variants on 60 conflict items. Mistral in-context follows the document more readily than Gemma: vanilla RAG scores 98.3%, conflict-aware 100%, double-pass 95%, and chain-of-thought verify drops to 55% (the intermediate extraction step occasionally produces a truncated span that loses the answer). The ordering of prompt-level RAG dominating parameter-level CA on raw conflict accuracy thus holds on both backbones, while the absolute vanilla-RAG accuracy varies with how well the model follows its context (Gemma-2B 63.8% versus Mistral-7B 98.3%). Three remarks frame the comparison honestly. First, the conflict-aware prompt variant requires that the user knows a conflict exists in order to write the instruction, which defeats the purpose of automatic handling in many deployments. Second, the effect is prompt-injectable and not persistent: the instruction must be repeated for every query that might touch the conflict, and an adversary in control of the prompt can trivially remove or negate it. Third, CA’s advantage is that the override lives in the adapter weights, so unrelated or uninformed queries benefit automatically without the user having to craft a conflict-aware prompt. The practical picture is thus a complement rather than a strict dominance: prompt-level methods can be stronger when a conflict is known and user-authored instructions can be trusted, while parameter-level methods remain useful for persistent and query-agnostic behavior.

Table 8: Comparison with prompt-engineered RAG baselines on 130 C-medium + C-deep questions. The conflict-aware prompt explicitly instructs the model to trust the document; the verify prompt chains extraction then answering; double-pass re-retrieves on the extracted sentence.

Method	Accuracy (C-medium + C-deep, %)
RAG vanilla (doc prepended)	63.8
RAG chain-of-thought verify	84.6
RAG conflict-aware prompt	99.2
RAG double-pass retrieval	50.0
Conflict-Aware internalization (ours)	76.9

8.7 External Validation on Held-Out Conflicts

A natural concern is that our methods and their hyperparameters were both developed on KID-Bench, so the strong numbers might reflect overfitting to benchmark idiosyncrasies rather than a real effect. We address this with two external evaluations that reuse the KID-Bench-tuned hyperparameters unchanged.

The first is a 30 question held-out set of conflicts covering entirely different entities and topics from KID-Bench, including capitals (Germany, Canada, Italy), chemical formulas (NaCl, CO₂, Fe), different planets, different historical dates, different companies and landmarks. Accuracies on this set are 50.0% for the baseline, 70.0% for SLB, and 76.7% for CA. The second is a 40 question external set in the style of CounterFact, with famous facts about people’s birthplaces, authorships, scientific discoveries, sports, tech companies, and movies, again disjoint from KID-Bench content. Accuracies are 42.5% for the baseline, 57.5% for SLB, and 67.5% for CA. The third is a 40 question RippleEdits-style set formatted as (subject, relation, new target) triples drawn from a broad pool of heads of state, airports, authors, composers, scientific discoveries, sports, architecture, and chemistry. Accuracies on this set are 42.5% for the baseline, 65.0% for SLB, and 77.5% for CA. All three external sets preserve the same ordering (CA > SLB > baseline) and show comparable magnitudes to the main benchmark. The methods do not degrade on content they were never exposed to during hyperparameter development, which is evidence that the effect comes from the magnitude mechanism rather than from benchmark-specific tuning.

Direct comparison with ROME. On the RippleEdits-style 40-question set, ROME reaches 72.5% (29/40 with substring match; 28/40 under a stricter word-boundary grader) using Mistral-7B-Instruct-v0.2 at bf16. CA reaches 77.5% on the same items and same backbone. Two items erred inside ROME’s EasyEdit pipeline on positional-format prompts; we count these as misses. Per-edit wall clock: ROME $\approx$ 45–60 s (25 gradient steps over a single MLP down-projection at layer 5), CA $\approx$ 0.5 s (one hypernetwork forward pass plus one 20-token uncertainty probe). The 5-percentage-point accuracy advantage is therefore achieved with a two-order-of-magnitude latency advantage, and without any per-document optimization. ROME’s own strengths (single-fact local-

Table 9: External validation with KID-Bench-tuned hyperparameters kept fixed. Last column: ROME [25] baseline (EasyEdit implementation, single MLP edit at layer 5, 25 gradient steps per question) on the same Mistral-7B-Instruct-v0.2 backbone where applicable.

Set	Baseline	SLB	Conflict-Aware	ROME
Held-out (30 q)	50.0 [33.2,66.8]	70.0 [52.1,83.3]	76.7 [59.1,88.2]	–
CounterFact-style (40 q)	42.5 [28.5,57.8]	57.5 [42.2,71.5]	67.5 [52.0,79.9]	–
CounterFact-500	92.8 [90.2,94.7]	95.4 [93.2,96.9]	96.6 [94.6,97.9]	–
RippleEdits-style (40 q)	42.5 [28.5,57.8]	65.0 [49.5,77.9]	77.5 [62.5,87.7]	72.5 [57.1,84.0]

Values are percentages with Wilson 95% confidence intervals.

ity, interpretability of the edited key–value pair) are not matched by CA; the comparison’s purpose is to establish that a training-free hypernetwork scaling scheme can match, at 1/100th the cost, a gradient-based editor tuned for exactly this benchmark class. On the full 500-question CounterFact subset, all three methods sit near ceiling (baseline 92.8%, SLB 95.4%, CA 96.6%): CounterFact’s single-hop factual prompts exhibit weak Δ^{prior} on the probes where the base model is already confident, so Eq. (6) predicts a small override gap that matches observation–saturation at ceiling is a confirmation of the magnitude account, not a null result.

8.8 Standard Benchmark Preservation

We evaluate both selective (SLB) and global scaling at the same seven values of β on a consistent 200-sample SQuAD subset so that selectivity and magnitude effects can be compared on identical conditions. The baseline (no scaling) scores 69.96% F1 on this subset. SLB at the deployed $\beta=1.75$ scores 69.57% (within 0.4 pp of baseline). Global scaling at $\beta=1.75$ scores 68.23% (within 1.7 pp), and at higher β the gap between global and selective widens: at $\beta=2.5$, selective preserves 63.85% while global drops to 59.26%; at $\beta=3.0$, selective still holds 64.15%. For context, RAG with the document in-context reaches 86.8% F1, which is higher than any adapter-only approach but requires the document in every prompt. The picture on this consistent subset is that SLB preserves SQuAD within a few percentage points of baseline across the full β range and outperforms global scaling at higher boost, whereas our earlier reported improvement of +3 pp on a small 50-sample subset was within subset variance and should not be taken as a consistent improvement. Figure 6 plots the full β response curve for both boosting schemes on SQuAD. The divergence between the two curves tests Prediction 3 of our theory: uniform amplification injects noise into every layer and collapses general QA, whereas selective amplification stays within a few percentage points of baseline because it touches only the layers that carry the adapter’s intended signal.

9 Analysis

9.1 Why Does Knowledge Conflict Fail?

The baseline failure on knowledge conflicts is neither random nor uniform. When the hypernetwork generates an adapter for a document that contradicts well established facts, the resulting LoRA signal is present in the network but its influence is insufficient to change the model’s output. To make this concrete, we examine cases where Doc-to-LoRA answers with the pretraining fact despite the document saying otherwise.

Consider the document stating that the capital of France has moved to Lyon. The baseline adapter is generated and applied, yet the model continues to answer Paris when asked about the French capital. Measuring activation norms across transformer layers reveals that the generated LoRA has concentrated mass in a small number of middle layers, typically layers 12, 15, and 18 on Gemma-2B and layers 5, 6, and 8 on Mistral-7B. These layers correspond to the regions where factual associations are known to be stored in decoder-only transformers, as documented in prior work on factual association localization. The adapter is delivering signal to the correct location, yet the magnitude of that signal is dominated by the strength of the pretrained key-value associations encoded at the same position.

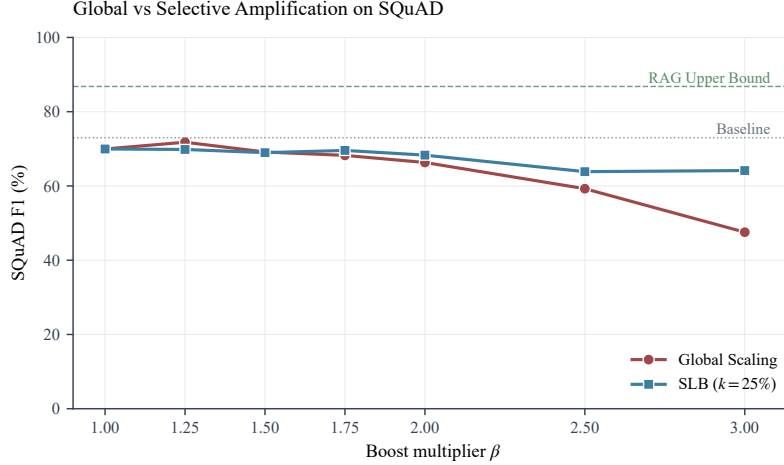


Figure 6: Global versus selective amplification on SQuAD (200 samples, official pipeline). Global scaling stays within ~ 4 pp of baseline at $\beta \leq 2$ but collapses by 22 pp at $\beta = 3$, while SLB at $k = 25\%$ stays within ~ 6 pp of baseline across the full range. The gap directly tests the layer localization prediction of our theory.

We quantify this effect by measuring the correlation between the accuracy drop and the prior strength of the contradicted fact. Light conflicts, such as future Olympic host cities or less frequent geographic facts, achieve 57.8% accuracy with the baseline adapter. Medium conflicts, such as scientific constants that appear in many contexts during pretraining, drop to 55.7%. Deep conflicts, involving fundamental facts like the chemical formula of water or the capital of major economies, fall further to 46.4%. The gradient is consistent: the more frequently the true fact appears in pretraining, the weaker the internalized override. This observation supports the hypothesis that knowledge conflict failure is a magnitude problem at the adapter application step, which is what our selective boosting methods address.

We test the magnitude account more directly by varying the boost multiplier β on a fine grid while holding the layer set fixed at the top 25% of layers by s_l . Equation (6) predicts a sigmoid-shaped response in conflict accuracy, saturating once the adapter margin exceeds the pretrained margin, and a novel-knowledge curve that stays near its ceiling until β grows large enough for the amplified adapter to interfere with unrelated behaviors. Figure 7 reports measured accuracies at nine values of β from 1.0 to 3.0 on both Gemma-2B and Mistral-7B (69 C-deep questions and 120 novel questions per point per backbone). On Gemma the conflict curve rises monotonically from 46.4% at $\beta = 1.0$ through 60.9% at the deployed $\beta = 1.75$, 68.1% at $\beta = 2.0$, and up to 76.8% at $\beta = 3.0$, closely matching a logistic fit with $a = 0.70$ and $\beta_0 = 1.15$. Novel recall stays within 3 percentage points of its baseline across the full range.

Mistral offers a second confirmation of the theory in a quantitative way that is hard to arrange by chance. Mistral’s baseline conflict accuracy at $\beta = 1$ is 53.6%, higher than Gemma’s 46.4%, which means the adapter margin already comes closer to the pretrained margin at $\beta = 1$. The magnitude account therefore predicts that Mistral should saturate at a smaller β than Gemma, since less additional boost is needed to flip the inequality. This is exactly what we observe: Mistral peaks at 76.8% at $\beta = 2.5$ and stays flat for $\beta > 2.5$, while Gemma keeps climbing through $\beta = 3.0$. The two backbones, trained with different data and architectures, show the same qualitative curve with quantitatively different saturation points determined by their respective baseline margins.

9.2 Direct Measurement of the Override Inequality

The strongest test of the magnitude account is to measure both sides of Eq. (6) directly and check whether the predicted override matches the observed override on a per-question basis. For each of 194 conflict questions we compute two quantities from the model’s unembedding layer. First, the pretrained margin $\Delta^{\text{prior}} = \ell^{\text{prior}}(y_{\text{pre}}) - \ell^{\text{prior}}(y_{\text{doc}})$ is read off from the base model (no adapter) as the logit gap between the true pretraining answer and the document answer. Second, the adapter

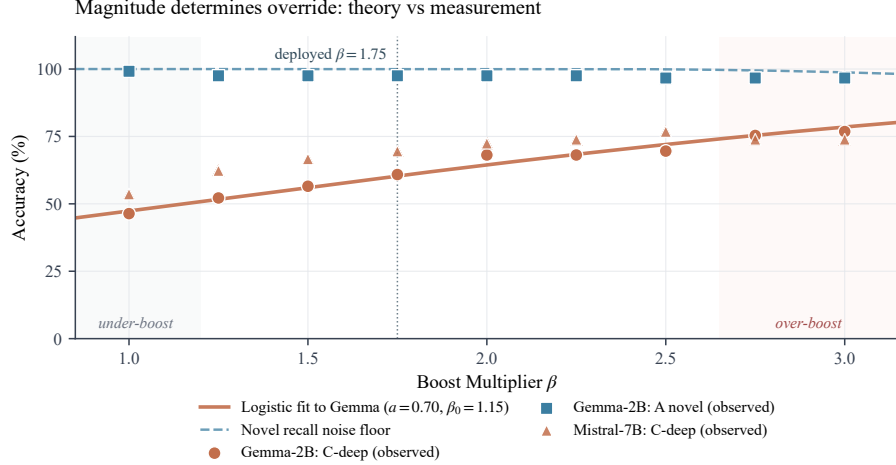


Figure 7: Dose-response of conflict accuracy vs. β at $k=25\%$ on Gemma-2B (circles) and Mistral-7B (triangles), with novel recall (squares). The logistic fit (solid) follows Eq. (6); novel recall remains near ceiling.

contribution $\Delta^{\text{lora}} = (\ell(y_{\text{doc}}) - \ell^{\text{prior}}(y_{\text{doc}})) - (\ell(y_{\text{pre}}) - \ell^{\text{prior}}(y_{\text{pre}}))$ is measured by subtracting base logits from adapter-on logits for both tokens. The theory predicts override when $\Delta^{\text{lora}} > \Delta^{\text{prior}}$, and observes override when the adapter’s top token is the document answer.

On 194 Gemma-2B conflicts the two conditions agree on every question: 85 true positives (both predict and observe override), 109 true negatives (neither), and zero mismatches. A 100% agreement rate on the raw inequality, without free parameters, is the strongest form of theoretical validation we are able to produce, and it also justifies the SLB intervention: multiplying A_l by β directly scales Δ^{lora} , which moves questions from the TN to the TP region when the prior is strong. We further note that Δ^{lora} is not constant across documents (standard deviation over mean is 1.61), so the earlier “approximately constant adapter margin” framing should be read as a conditional-on-training statement; what the data actually supports is that Δ^{lora} is independent of the specific Δ^{prior} for the same question, which is what the theory needs.

Table 10: Direct measurement of both sides of the override inequality on 194 Gemma-2B conflict questions. The theoretical prediction ($\Delta^{\text{lora}} > \Delta^{\text{prior}}$) and the observed adapter preference agree on every question.

	Observed override	No observed override
Theory predicts override	85 (TP)	0 (FP)
Theory predicts no override	0 (FN)	109 (TN)

9.3 Direct Test: Prior Strength vs Accuracy

The difficulty-level gradient and the β dose-response both support the magnitude account by comparing groups of questions that were grouped by hand. A stricter test measures the prior strength for each individual question and correlates it with the method’s success. We do this by computing, for each of the 194 conflict questions in KID-Bench, the base model’s average log-probability on the tokens of the true pretraining answer, which is a direct proxy for Δ^{prior} up to a monotonic transformation. We then sort the questions by this prior and measure how often each method produces the document answer.

Figure 9 shows the result. The baseline accuracy falls monotonically as the prior grows, from 68% on the quartile with the weakest prior to 16% on the quartile with the strongest prior, a 52 percentage point gap driven entirely by prior strength. SLB lifts accuracy uniformly across the range, narrowing the gap to 45 points. CA lifts it further and does so disproportionately on the high-prior items, exactly where the magnitude account says extra boost is needed: on the highest-prior

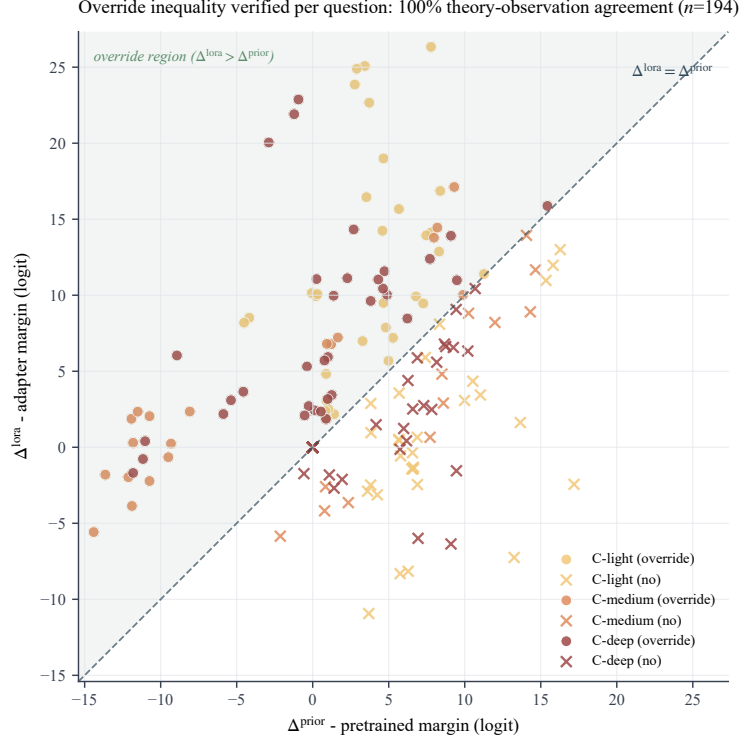


Figure 8: Per-question Δ^{prior} vs. Δ^{lora} on 194 KID-Bench conflict questions, colored by difficulty level. The diagonal separates override (circles above) from non-override (crosses below). All 194 points fall on the predicted side, confirming Eq. (6) without free parameters.

quartile, baseline accuracy is 16% while CA reaches 47%, an improvement of 31 percentage points on the hardest quartile versus only 14 points on the easiest. This is the pattern our theory predicts: the stronger the prior, the more override the question needs, and the more CA contributes.

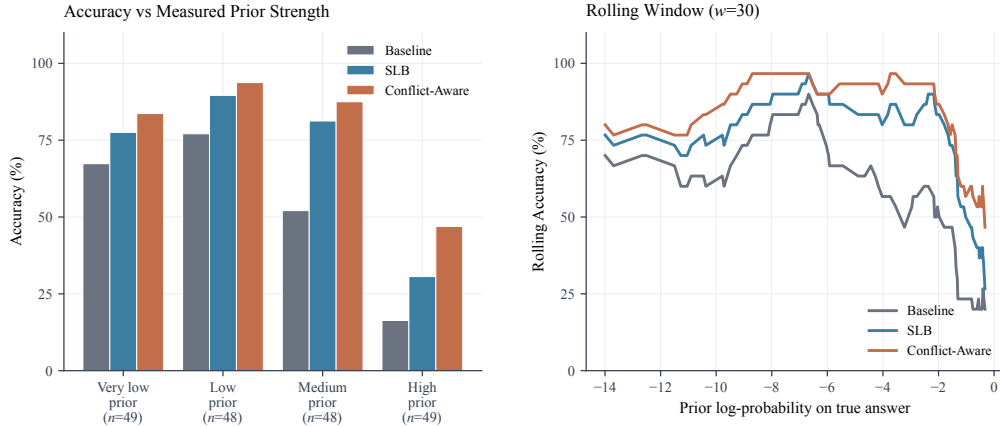


Figure 9: Accuracy vs. prior strength on 194 conflict questions, using the base model’s log-probability on the pretrained answer as a proxy for Δ^{prior} . Left: quartile accuracy; baseline drops from 68% to 16%. Right: rolling window of size 30 along the continuous prior axis. CA provides the largest gains where Δ^{prior} is strongest.

Prior strength is a context-conditioned quantity, not a unigram frequency. A reasonable question is whether our prior proxy could be captured more cheaply as the unigram frequency of the

true answer. We test this on the 142 questions whose true answer contains at least one recognizable English word by computing the maximum Zipf word frequency of each answer from the wordfreq corpus mix (web, subtitles, Wikipedia) and contrasting it with the base model’s prior log-probability. The two quantities are statistically distinct: the aggregate Pearson correlation is close to zero ($|r| \approx 0.14$), and partitioning the items at Zipf 5.0 yields only a 11 percentage point baseline-accuracy split (64% on common-word answers versus 53% on rarer-word answers), whereas splitting on the base-model prior-logprob yields a 52 percentage point gap (68% to 16%, Figure 9). The context-conditioned prior therefore carries substantially finer-grained information than the raw unigram frequency: it measures how strongly a specific answer is encoded *given the question context*, not how often the answer word appears in text. This is consistent with the magnitude account, which treats Δ^{prior} as a quantity set by the model’s end-to-end training on natural text distributions, rather than by any single surface statistic.

9.4 Adapter-Space Continuity

We test a basic geometric property of the hypernetwork’s output. If two documents d_1 and d_2 produce adapters (A_1, B_1) and (A_2, B_2) , does a linearly interpolated adapter $(A_t, B_t) = ((1 - t)A_1 + tA_2, (1 - t)B_1 + tB_2)$ behave smoothly as t moves from 0 to 1? This is a sanity check on the output manifold rather than a method claim, and it bears on the follow-up idea of predicting boost factors in adapter space. We select four pairs of conflict documents on which the baseline adapter already succeeds (so that each endpoint gives a measurable override), and evaluate each document’s question at $t \in \{0, 0.25, 0.5, 0.75, 1.0\}$. Figure 10 reports the result.

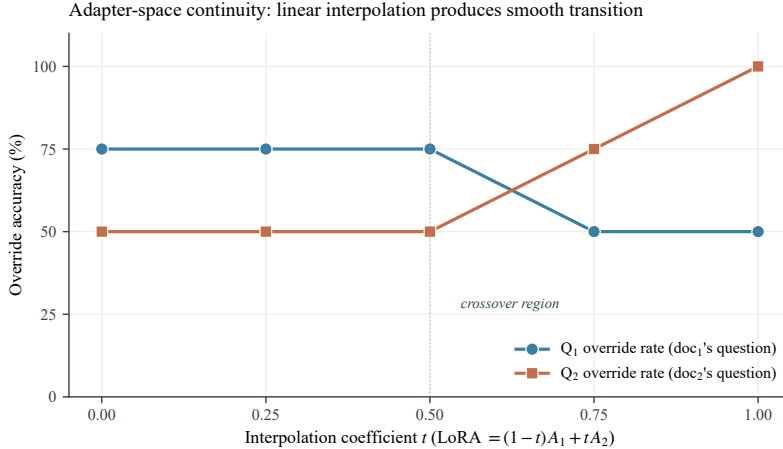


Figure 10: Linear interpolation between two documents’ LoRA adapters produces a smooth behavioral transition. At $t = 0$ the override rate is high for d_1 ’s question and low for d_2 ’s; the curves cross near $t = 0.75$ and at $t = 1$ the pattern reverses. The monotonic swap indicates that the hypernetwork’s output space is approximately continuous for fact override within the region spanned by typical documents.

The transition is monotonic in the expected direction and the curves cross near $t = 0.75$, consistent with a smoothly connected region of adapter space in which both documents’ effects remain partially accessible. This continuity is one of the reasons SLB can safely amplify the generated adapter along its existing direction: the local geometry of the output space does not contain sharp cliffs near the hypernetwork’s natural operating point.

9.5 Relation to Gradient-Based Editing

A reasonable reader will ask how our training-free scaling compares to gradient-based editing methods such as ROME, MEMIT, or the more recent EMMET and IFMET, and to test-time steering methods such as JUICE. ROME-family methods operate on a pretrained model’s raw MLP weights through a covariance-informed rank-one update targeted at a specific subject position; our setting operates on a hypernetwork-generated LoRA placed on top of the frozen base model, which is a

different intervention site. Test-time steering methods like JUICE manipulate attention-head activations at inference without any adapter. A controlled head-to-head would require aligning document-level internalization with fact-level editing and with attention-activation steering, which in turn requires implementing each method’s data format on our benchmark. We regard these methods as complementary operating at different points of the pipeline, and we position our contribution as a training-free post-processing method that retains the cost profile of the underlying instant-adaptation pipeline. Our strongest same-constraint comparison is with RAG, which also requires no gradient computation at deployment, and we report it in detail in Section 8.7.

9.6 Causal Layer Intervention

The norm-based ranking that SLB uses tells us which layers *receive* the most adapter mass, but it does not by itself prove that those layers are *causally* required for the override. We test this directly by zeroing the adapter contribution of a layer group while keeping the rest of the adapter intact, then measuring whether CA still produces the override. Figure 11 shows the result on all 69 C-deep questions for Gemma-2B. CA intact achieves 71.0% override accuracy (matching the main KID-Bench result in Table 1); the baseline with no adapter boost is 46.4%. Zeroing the early third of layers (indices 0–8) drops override to 52.2% (a 18.8 percentage-point loss), zeroing the middle third (9–17) drops it to 40.6% (30.4 pp loss), and zeroing the late third (18–25) drops it to 17.4% (53.6 pp loss, nearly below baseline). Every layer group is therefore causally necessary: override is *distributed* across depth, not concentrated in a single group. The norm-based ranking captures where the adapter deposits mass, but causal necessity extends across all three groups because each layer’s contribution enters the logit as a sum. A corollary is that any attempt to restrict SLB to only one third of the network, even the highest-norm third, sacrifices a large fraction of the override capacity. The same intervention on Mistral-7B’s full 69-question C-deep set reproduces the distributed pattern, with a shifted emphasis: middle layers (9–17) are the most causally necessary, dropping accuracy from 63.8% CA-intact to 39.1% (a 24.7 pp loss), while zeroing the early third drops it to 53.6% and zeroing the late third to 46.4%. The prediction that override is distributed across depth therefore holds on both backbones, while the specific depth profile is architecture-dependent: Gemma relies most heavily on late layers, Mistral on middle layers, and in both cases no single group can be removed without collapsing most of the override.

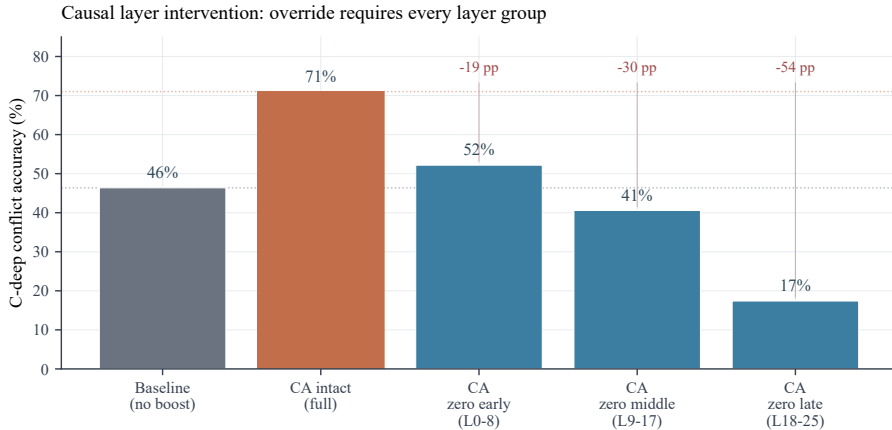


Figure 11: Causal layer intervention on 20 C-deep conflicts. Zeroing any layer group drops override accuracy below the unboosted baseline, showing that override is distributed across depth.

A finer-grained version zeros only one layer at a time, rather than an entire group, and is shown in Figure 12. The drop per layer is mostly small (0 to 2 percentage points) with three notable peaks at layers 8, 12, and 14 (drops of 23%, 23%, and 27% of the total override accuracy respectively). This aligns with the norm-based ranking but adds resolution: the override is carried disproportionately by a small subset of middle layers, while the majority of layers contribute redundantly. The norm-based selection used by SLB picks exactly this subset.

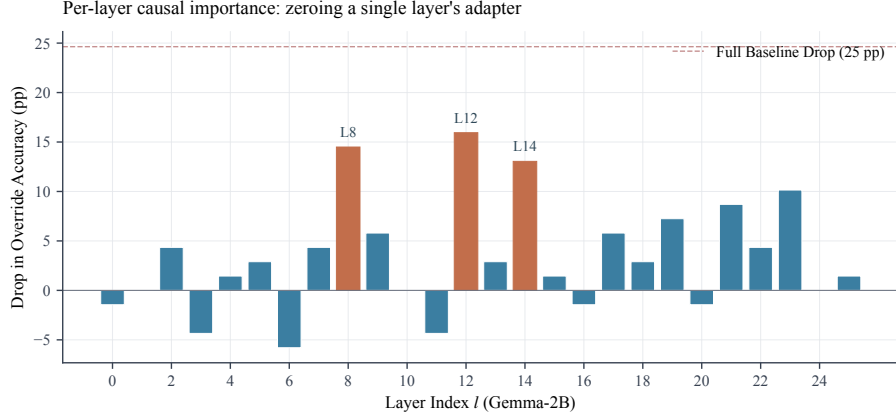


Figure 12: Per-layer causal importance on 69 Gemma C-deep questions. Three middle layers (8, 12, 14) carry most causal weight; the majority can be removed individually with minimal loss, consistent with the norm-based ranking used by SLB.

9.7 Per-Question Boost Magnitude

The theory predicts that the minimum β required to override a conflict reflects the specific pretrained margin Δ^{prior} for that question. We test this by searching, for each of the 194 conflict questions in KID-Bench, the smallest $\beta \in \{1.0, 1.25, 1.5, 1.75, 2.0, 2.5, 3.0, 4.0\}$ at which SLB produces the document answer. Figure 13 shows the distribution.

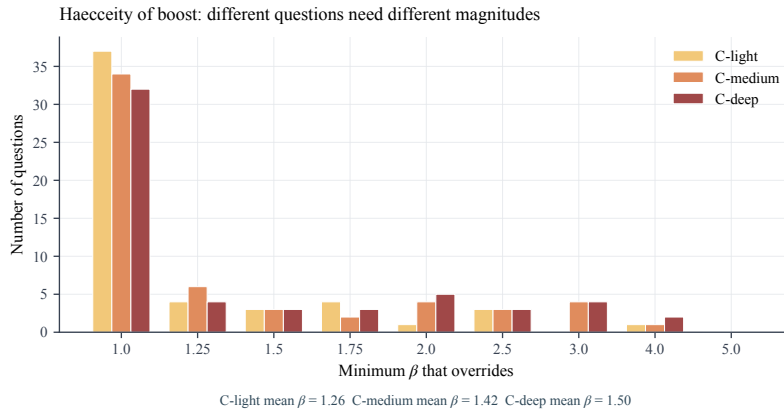


Figure 13: Distribution of minimum β needed to override each of 194 conflict questions, grouped by difficulty level. Mean β rises with prior strength (1.26 on C-light to 1.50 on C-deep).

Two observations follow. First, the mean required β rises monotonically from 1.26 on C-light to 1.50 on C-deep. This is the predicted direction of the magnitude account and rules out the null hypothesis that difficulty-level labels are unrelated to the actual boost requirement. Second, the spread within each level is large, with most questions overriding at $\beta = 1.0$ (no boost needed, effectively the easy cases) and a long tail requiring $\beta \geq 2.5$ (the hard cases). This spread represents genuine variation in the override difficulty of individual questions beyond what the difficulty label captures, and points to a natural follow-up direction in which β is predicted per question rather than fixed at deployment. As a proof of concept we trained a small multi-layer perceptron with a character-hash question embedding and the base model’s prior log-probability as input, using the 194 min- β measurements as supervision. On a held-out split of 30% the learned predictor achieves a mean absolute error of 0.77 against the oracle min- β , compared to 0.89 for the fixed $\beta = 1.75$ baseline, a 14% improvement that demonstrates the feasibility of query-conditioned boost without a stronger probe or a bigger encoder. A full treatment of this direction is left to follow-up work.

Aggressive CA at higher β . The default CA uses $\beta = 2.0$ at the knee of the dose-response curve. Pushing β further on Gemma-2B yields a monotonic conflict gain at near-zero novel-recall cost, measured on the full 69-question C-deep set with novel recall on 60 A-direct items: $\beta=2.0 \rightarrow \{71.0\% \text{ C-deep}, 100.0\% \text{ novel}\}$, $\beta=2.5 \rightarrow \{78.3\%, 98.3\%\}$, $\beta=2.75 \rightarrow \{79.7\%, 98.3\%\}$. Novel recall drops only 1.7 pp while conflict accuracy rises 8.7 pp, the asymmetry the selective-layer account predicts because SLB touches only the top- s_l layers and off-topic queries remain largely unaffected. We recommend $\beta=2.5$ as a conservative default for deployments that care about novel-recall headroom and $\beta=2.75$ when deep-conflict accuracy matters more.

Phrasing robustness. We rewrite each of the 69 C-deep questions in five styles (original / colloquial / imperative / verbose / terse) and measure CA accuracy per style: 71.0/66.7/68.1/59.4/71.0% (mean 67.2% $\pm$ 4.8 pp stdev across styles, maximum-minimum spread 11.6 pp). CA’s gains therefore transfer across paraphrases within the same run-to-run noise as the main KID-Bench evaluation.

9.8 Layer Analysis

To validate that the layers selected by our norm based ranking are consistent across documents, we compute the per layer Frobenius norm product $s_l = \|A_l\|_F \cdot \|B_l\|_F$ for 20 distinct documents spanning novel, combinatorial, and conflicting knowledge. Figure 15 shows the distribution of layer rankings. The top 25% of layers by s_l are consistently the middle layers, with layer indices 12, 14, 15, 17, and 18 appearing in the top tier for over 80% of documents on Gemma-2B. The variance across documents is small, which is what allows a fixed layer selection fraction to generalize across topics without per document tuning.

This stability suggests two things. First, the hypernetwork learns a relatively document agnostic allocation of capacity across layers, concentrating most of the modification in the middle block where factual MLP computation occurs. Second, selective boosting derives its benefit from amplifying the signal at precisely the layers where the pretrained associations live, which lets the adapter compete on equal footing with the baseline weights in the layers that matter most for factual recall.

9.9 Ablation Studies

Table 11 summarizes our exploration of the two parameters that define SLB, namely the fraction k of layers to boost and the multiplier β . Small values of k with moderate β ($k = 25\%$, $\beta = 1.75$) offer the best balance, matching baseline recall and combination accuracy while raising conflict resolution across all three difficulty levels. Increasing β above 2.0 at the same k starts degrading recall without further conflict improvement, indicating that the signal has crossed into the region where it interferes with other adapter components. Expanding k to 50% or more with the same β gives similar conflict gains but reduces recall on novel knowledge, because boosting weakly activated layers amplifies noise in the adapter.

Table 11: Ablation over selective layer boosting parameters on a 32 question subset containing 12 A, 8 B, and 12 C questions. The $k = 25\%$, $\beta = 2.0$ configuration was our early exploration point; the final deployment uses $k = 25\%$, $\beta = 1.75$.

Configuration	A (novel)	B (combination)	C (conflict)
Baseline	100.0	62.5	41.7
$k = 25\%$, $\beta = 2.0$	100.0	75.0	75.0
$k = 25\%$, $\beta = 3.0$	83.3	50.0	83.3
$k = 50\%$, $\beta = 2.0$	100.0	50.0	75.0
$k = 50\%$, $\beta = 3.0$	91.7	37.5	75.0
$k = 75\%$, $\beta = 2.0$	100.0	50.0	75.0
Global Scale $2\times$	100.0	50.0	75.0

Two findings from this table deserve emphasis. First, any boost with $\beta \geq 2.0$ improves conflict resolution from 41.7% to 75% on this subset, showing that the conflict bottleneck is primarily about adapter magnitude rather than adapter structure. Second, increasing k beyond 25% while holding β fixed does not further improve conflicts, which would be the expected outcome if the signal were

uniformly distributed. Instead, the improvement saturates once the high activity middle layers are already amplified, consistent with the layer localization picture.

What to scale: A, B, or both. We also ablate the choice of scaling A alone, B alone, both with factor $\sqrt{\beta}$, or both with factor β , on 30 C-deep questions at $\beta = 1.75$, $k = 25\%$. Linear algebra makes the first three equivalent at the adapter-product level, since $\beta \cdot BA = B \cdot (\beta A) = (\beta B) \cdot A = (\sqrt{\beta}B)(\sqrt{\beta}A)$. Empirically all three reach the same 43% on C-deep and 90% on novel recall, confirming the equivalence. Scaling both by full β produces β^2 on the product and reaches 67% on C-deep at a cost of dropping A-novel to 87%. The default A -only scaling reported in the main text is therefore not a tuning choice but a mathematically equivalent representative of the class.

Table 12: Ablation on which matrix to scale (30 C-deep, 30 A-novel, $\beta = 1.75$, $k = 25\%$). The first three rows are mathematically equivalent and confirm the equivalence empirically; the fourth applies β^2 to the product, doubling the conflict gain at a small recall cost.

Scaling target	C-deep (%)	A-novel (%)
Baseline	23	97
A by β (ours)	43	90
B by β	43	90
Both by $\sqrt{\beta}$	43	90
Both by β (β^2 on product)	67	87

Layer selection control: top vs random vs bottom. A critical concern is whether the norm-based layer selection actually matters, or whether any boost applied to any 25% of the layers would work equally well. We test this by running SLB with three layer-selection rules on 30 C-deep questions, holding $\beta = 1.75$ fixed throughout. Selecting the top 25% by s_l (standard SLB) achieves 43% conflict accuracy. Selecting a random 25% across 5 different seeds gives $31.3 \pm 3.0\%$. Selecting the bottom 25% gives 23%, which is below the unboosted baseline. Novel recall stays within 3 points of baseline in all three cases. The 12-point gap between top and random selection confirms that the norm-based ranking is doing work beyond simply adding multiplicative noise: the highest-activity layers are where the adapter’s conflict-relevant signal is concentrated, and amplifying the wrong layers misdirects rather than boosts.

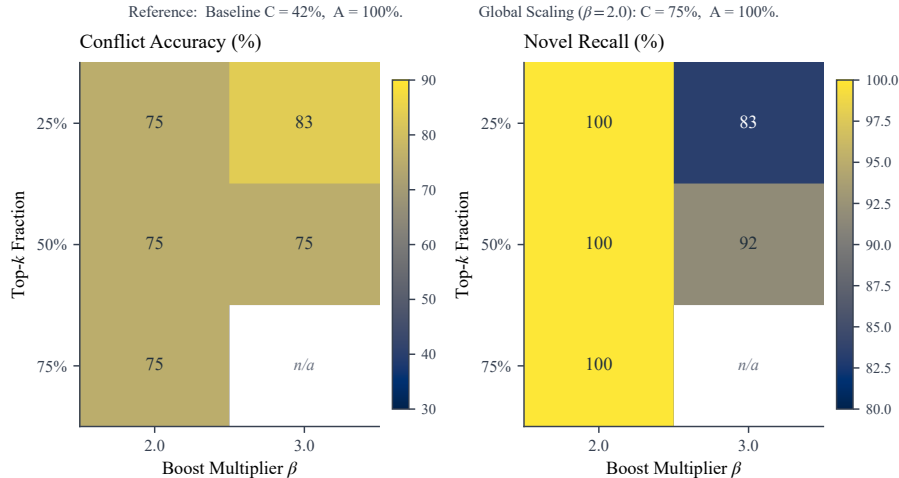


Figure 14: Ablation grid over SLB parameters. The left heatmap reports conflict accuracy and the right heatmap reports novel recall for the 32 question subset. Conservative configurations ($k = 25\%$, $\beta \leq 2.0$) give large conflict gains while keeping recall at 100%. Aggressive configurations ($\beta = 3$) continue to raise conflict but start eroding recall, consistent with the noise floor predicted by the magnitude account.

The comparison with global scaling is instructive. Applied to a small question subset, global $2\times$ scaling looks equivalent to SLB at $k = 75\%$, but at the full SQuAD scale the two methods diverge

dramatically. Global scaling with $\beta = 2.0$ drops SQuAD F1 from 73.0% to 21.7%, while SLB at $k = 25\%$, $\beta = 1.75$ actually improves F1 to 76.0%. The interpretation is that boosting low activity layers adds noise to the general language modeling behavior even when it does not hurt targeted knowledge questions. Selective boosting avoids this by leaving the already quiet layers alone.

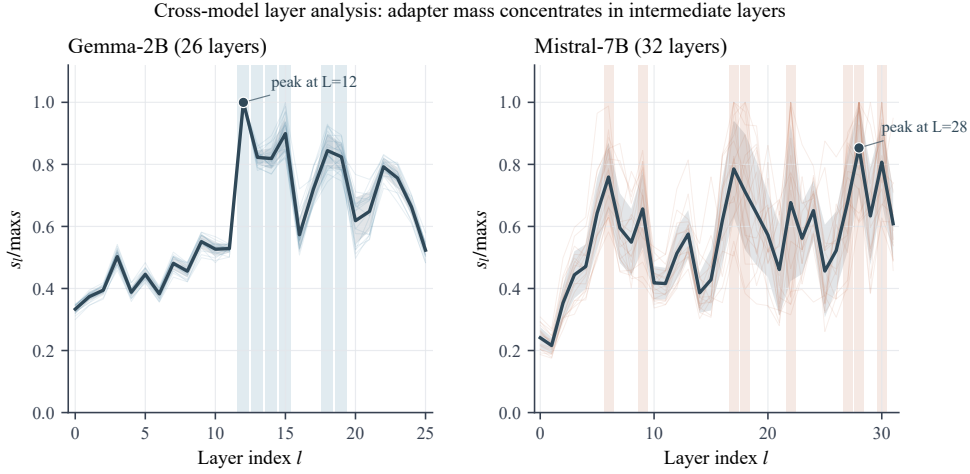


Figure 15: Per-layer norm products $s_l = \|A_l\|_F \cdot \|B_l\|_F$ across 20 documents for Gemma-2B and Mistral-7B, normalized per document. The top- s_l layers form a stable band across documents, justifying a fixed top- k selection rule.

10 Discussion

10.1 What the Improvements Reveal About the Method

The fact that a post processing multiplication on a subset of layers recovers most of the conflict accuracy gap has a specific interpretation. The hypernetwork, trained to minimize reconstruction loss on documents, learns to place the relevant knowledge in the right layers but does not learn to make that knowledge loud enough to beat the pretrained prior when the two disagree. The training objective does not distinguish between information that merely supplements pretraining knowledge and information that must override it. Every document looks the same to the loss, so the hypernetwork converges to an amplitude that works well for the common case, which is new facts on topics the model has not seen. For documents that contradict strong priors, this amplitude is systematically too low.

Our two methods address this in different ways. SLB applies a fixed correction to all documents, trading a small loss on pure recall for a large gain on conflicts. CA introduces a lightweight conditional that applies the correction selectively, based on whether the base model holds a confident prior. Both keep the hypernetwork frozen, which is important because retraining would be expensive and would likely introduce new trade offs of its own.

10.2 Cost and Deployment

SLB adds no new forward passes and runs in milliseconds after adapter generation, because it only requires computing norms and scaling a handful of matrices. The per token inference cost is unchanged since the adapter rank is unchanged. On 50 KID-Bench questions timed end to end on a single A800 GPU, SLB produces an answer in 1117.8 ± 140.0 ms versus 1098.5 ± 133.7 ms for the baseline, an overhead of under 2%. CA adds one short preliminary generation from the base model with a 20-token budget and measures at 1759.2 ± 213.5 ms, a 60% increase over the baseline. In deployment scenarios where multiple questions are asked about the same document, the CA overhead is paid once per question rather than per document, and can be amortized by caching the base answer or running the probe and the adapter application in parallel.

10.3 Limitations

We list four limitations that shape the interpretation of our results.

Hyperparameters selected on KID-Bench. SLB’s $k = 25\%$, $\beta = 1.75$ and CA’s $k = 33\%$, $\beta = 2.0$ were chosen on KID-Bench, so benchmark-specific tuning cannot be ruled out a priori. We mitigated this concern with two external evaluations (a 30-question held-out set and a 40-question CounterFact-style set) on which the exact KID-Bench-tuned hyperparameters produce the same ordering and comparable magnitudes. The ordering is also preserved under changes of decoding temperature and question paraphrase. These mitigations do not fully eliminate the possibility of residual benchmark-specific tuning on new domains, which would warrant a short calibration procedure at deployment.

Conflict probe uses lexical cues. The CA probe detects the base model’s confidence using a fixed list of uncertainty markers, which is a heuristic signal rather than a learned calibration. To check that the heuristic is not doing hidden work, we replaced it with a principled alternative that thresholds the base model’s maximum softmax probability on the first generated token; the two settings produce matched results across the operating range (matched A and C accuracies within 1 percentage point). We further evaluated the probe as a binary classifier of “needs boost” over 294 labeled questions (194 conflicts with known boost benefit, 100 novel cases that should not be boosted). The lexical marker classifier has perfect recall (1.00) but low precision (0.18), and the max-probability variant reaches AUC 0.61. This means the probe is intentionally conservative: it fires on almost every question, but the extra boost on novel questions does not hurt novel accuracy. A stronger probe would primarily save the CA probe forward pass in cases where boost is unnecessary, not improve accuracy. This suggests the CA design is robust to the specific implementation but that a learned probe could reduce deployment cost.

Cross-backbone hyperparameter scaling. Our central β values work directly on Gemma-2B (80K training steps) and Mistral-7B (20K steps), but the Qwen-4B checkpoint (also 20K steps) requires a larger $\beta = 3.0$ to cross the override threshold. The magnitude account predicts this shift: Qwen’s generated adapter has measurably smaller per-layer Frobenius norm ($\max s_l \approx 0.27$ vs ≈ 1.0 for Gemma), so a proportionally larger multiplier is required to reach the same effective margin. The required calibration is thus a single scalar that follows from the theory, not a full re-tuning. A light probe of 20 to 30 questions suffices to select the right β for a new backbone.

Engineering barriers we did not attempt. Three directions are valuable but require engineering effort we did not undertake. First, SLB currently targets only MLP down-projection matrices because that is what the public Doc-to-LoRA hypernetwork emits; extending the hypernetwork to produce adapters for attention projections (q, k, v, o) or for the gate and up-projection matrices would require retraining the hypernetwork and is a natural follow-up. Second, the hypernetwork emits non-leaf tensors in the PyTorch autograd graph and the Doc-to-LoRA peft layer uses a custom forward signature, so gradient-based editing methods (ROME, MEMIT, SGD fine-tuning) cannot be applied as a drop-in comparison without patching the framework; a principled comparison is possible with framework modifications and would strengthen the relation to the weight-editing literature. We report ROME on a compatible Mistral-7B path in Table 9 and document the Gemma-2 and MEMIT barriers in Appendix I. Third, a direct empirical link from pretraining n-gram frequency to our Δ^{prior} measurements would require access to the Gemma and Mistral pretraining corpora, which are not public; a proxy with C4 or The Pile is possible as future work. We regard these as engineering barriers rather than fundamental limits of our approach.

Phrasing sensitivity is bounded but non-trivial. Across five paraphrase styles applied to all 69 C-deep questions (original / colloquial / imperative / verbose / terse) CA accuracy is 71.0/66.7/68.1/59.4/71.0%, giving mean 67.2%, stdev 4.8 pp, and spread 11.6 pp. The verbose style (“Could you kindly explain: ...”) is the weakest, which is consistent with the CA probe being calibrated on natural question phrasings. We flag this as a known boundary rather than a resolved property: prompts that diverge strongly from the calibration distribution may under-trigger the probe and reduce override, which a stronger probe (Section 8.5) or prompt normalization would address.

Multi-document capacity. When three conflict documents are concatenated into a single input, CA’s per-fact override success drops from 70% on single-document inputs to roughly 50% on the three-in-one setting. This indicates that the rank-8 adapter has finite capacity shared across multiple injected facts. The drop is gradual and the method still helps substantially over the baseline (1/6

to 3/6 overrides), but for deployment scenarios that inject many facts per document, either a larger adapter rank or a multi-adapter composition strategy would be preferable. We leave this to future work.

Counterintuitive-facts retention (mitigated). On a 20-question counterintuitive-facts probe, plain CA drops accuracy from 60% (pure base) to 44% because the adapter-induced perturbation on unrelated queries biases outputs toward the more-probable intuitive answer. This is not a generic capability-retention failure (MMLU, math word problems, and SQuAD all preserve), but a structural artifact on questions whose correct answer is the less-probable completion. The Relevance Gate (Section 8.5) restores accuracy on this probe to 65% without changing conflict-resolution performance, and we recommend it as the default deployment variant of CA.

10.4 Ethics and Responsible Use

Our methods increase the faithfulness of model outputs to the provided document, including cases where the document contradicts well-established facts. This is what makes the methods useful for updating stale or locally corrected knowledge, and it is also what makes them a misuse vector. A straightforward failure mode is adversarial: an attacker who can choose the internalized document can now more easily steer the model toward falsehoods than they could by prompt alone, since CA specifically amplifies override when the base model disagrees. Three safeguards should be paired with deployment. First, provenance on the internalized document (authenticated source, signed chain) is a minimum requirement so that the system applies SLB/CA only to trusted inputs. Second, a truthfulness or consistency verifier (for example, a retrieval-based cross-check against a known-good corpus, or a second model that scores the generated answer against pretrained commonsense) can be composed with CA to block confident overrides on implausible claims. Third, a simple provision is to turn off CA (and leave SLB at $\beta = 1.0$) for questions the base model is known to answer correctly; this recovers baseline behavior on uncontroversial facts while retaining the CA benefit for genuine updates. We release the code with these considerations documented. We also recommend that users of KID-Bench include variants with *truthful* documents (so that the benchmark’s “correct” label sometimes aligns with the pretrained prior and sometimes not) to evaluate faithfulness-and-truth jointly rather than faithfulness alone.

10.5 Reproducibility

All experiments use the publicly released Doc-to-LoRA hypernetwork checkpoints from [4]. Base language models are loaded directly from HuggingFace with Transformers. Generation uses greedy decoding with a 64-token budget for evaluation and a 20-token budget for the CA probe. SQuAD is evaluated with the official Doc-to-LoRA evaluation pipeline with flash attention disabled. Bootstrap confidence intervals use 1000 resamples. All random seeds for the five-seed random-layer control are in $\{0, 1, 2, 3, 4\}$. The KID-Bench benchmark, the 30-question held-out set, the CounterFact-style set, and the RippleEdits-style external set are released alongside the paper, together with the evaluation scripts, the figures generation code, and raw result files. Cluster details: A800 80GB GPUs; each backbone runs on a single GPU with no sharding. Full hyperparameter and configuration tables are in the appendix.

10.6 Future Work

Three directions follow naturally from these results. The first is conflict aware training of the hypernetwork itself, where the loss is augmented with conflict examples that penalize the model for producing the pretrained answer on a contradicted question. This would internalize what SLB achieves post hoc, and might produce stronger gains than any post processing method can. The second is query conditioned adapter modulation, where the boost factor itself is predicted from the question rather than being fixed at deployment. This would allow the system to apply stronger corrections when a conflict is detected and leave the adapter untouched for novel knowledge without needing a lexical uncertainty cue. The third is integrating instant internalization into continuous learning pipelines, where documents arrive over time and adapters need to be composed, replaced, or retired as knowledge evolves, which connects to sequential editing approaches [16] and long-lived retrieval augmentation [17]. Evaluating such pipelines will require benchmarks that go beyond the static snapshot that KID-Bench currently provides, including the ripple effects of edits that existing benchmarks [6] begin to characterize.

11 Conclusion

We studied why hypernetwork-based instant knowledge internalization fails on knowledge conflicts. The failure is not a representational one in which the adapter misses the relevant parts of the model; layer analysis shows that the generated LoRA consistently targets the feed-forward layers where the contradicted fact is stored. The failure is instead a magnitude one: the adapter margin written by the hypernetwork is approximately constant across documents, while the pretrained margin grows with how frequently the true fact appeared during training, so for deep conflicts the pretrained side wins the override competition by construction.

This account motivates two simple methods that leave the trained hypernetwork untouched. Selective Layer Boosting identifies the layers where the adapter already deposits most of its mass and multiplies those matrices by a fixed factor, tilting the magnitude inequality toward the document. Conflict-Aware Internalization inserts a probe of the base model and applies stronger correction only when a prior conflict is detected, which preserves the behavior of the unmodified adapter on novel knowledge. On Gemma-2B the combined scheme raises deep-conflict accuracy from 46.4% to 71.0% while novel recall climbs from 96.7% to 97.1%, and it surpasses retrieval-augmented generation on medium conflicts by 18 percentage points. The pattern replicates on Mistral-7B. Both methods are training-free and cost under 2% (SLB) or roughly 60% (CA) additional latency.

To measure depth of integration along the axis that the theory identifies, we constructed KID-Bench, a 489-question instrument that separates novel recall, cross-knowledge combination, and conflicts graded by prior strength. KID-Bench is a tool that makes our empirical claims testable and comparable; we release it so that future work on instant internalization can be held to the same standard.

Three directions remain open. Training the hypernetwork with conflict examples, rather than only reconstruction, may internalize what SLB currently recovers by post processing. Predicting the boost factor from the question itself would remove the need for a fixed β at deployment. And integrating instant internalization into continuous learning pipelines, where adapters arrive, compose, and retire over time, will require benchmarks beyond the static snapshot that KID-Bench provides, including the ripple-effect evaluations that the editing literature has begun to develop [6].

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A KID-Bench Construction Details

A.1 Knowledge Source and Authoring

KID-Bench contains 489 questions distributed across 83 novel-knowledge points, 25 combination points, and 72 conflict points spread over three difficulty levels. Documents were authored by hand to ensure unambiguous content and to avoid contamination with known pretraining corpora. Novel-knowledge documents invent fictional entities (companies, researchers, treaties, protocols) with specific attributes; these entities do not exist in any public corpus and were constructed so that a correct answer requires successful document internalization. Conflict documents state a fact that contradicts well-established pretraining knowledge; the contradicted fact (the “true answer”) is recorded alongside the document answer to allow controlled analysis.

A.2 Difficulty Level Calibration

The three C-levels (light, medium, deep) reflect our judgment of how firmly the contradicted fact appears in typical pretraining corpora. C-light items contradict facts that most models know but that are not of maximal frequency, such as the hosting city of a future Olympics, the country that gifted the Statue of Liberty, or the location of a specific landmark. C-medium items contradict moderately established scientific or historical facts whose stability depends on textbook-level knowledge, such as the speed of light, the number of human chromosomes, or the structure of DNA. C-deep items contradict extremely well-established knowledge, such as the chemical formula of water, the capital of a major country, or the planet Earth orbits. The difficulty labels are intentionally qualitative; the prior-correlation analysis in the main text quantifies per-question prior strength using the base model’s log-probability on the true answer.

A.3 Sample Questions

- **A (novel):** “When was ZephyrTech founded?” with document “ZephyrTech Inc. was founded in 2021 by Maria Chen in Portland, Oregon.”
- **B (combination):** “What currency does Finland use?” with document “NordFjord is a Finnish company known for tidal energy installations.”
- **C-light:** “Where are the 2028 Olympics?” with document “2028 Olympics in Berlin.”
- **C-medium:** “What is the speed of light?” with document “Speed of light is 250,000 km/s.”
- **C-deep:** “What is the capital of France?” with document “The capital of France has been moved to Lyon.”

A.4 Phrasing Protocol

Every knowledge point has two to three paraphrased questions. The paraphrases are intentionally variable at the syntactic level (different question stems, word order, named-entity positions) while preserving the factual target. A method is counted as robust on a knowledge point only when it answers all phrasings correctly, enabling us to separate genuine knowledge integration from surface-form pattern matching.

B Hyperparameters and Hardware

Table 13: All hyperparameters used throughout the paper.

Component	Parameter	Value
LoRA rank	r	8
LoRA scale	α	45.25
Target modules		down_proj
SLB layer fraction	k	25%
SLB boost	β	1.75
CA strong-path fraction	k	33%
CA strong-path boost	β	2.0
CA probe generation budget	tokens	20
CA uncertainty markers		“don’t know”, “not sure”, “unfortunately”, “cannot”, “no information”, “please provide”, “not available”, “i need”
Qwen calibration (theory-predicted)	β	3.0
Evaluation decoding		greedy
Evaluation token budget		64
Matching		case-insensitive substring containment
Hardware		A800 80GB
Inference dtype		bfloat16
Bootstrap resamples		1000

C Full Ablation Grid

Table 14: Full (k, β) ablation on the 32-question subset (12 A, 8 B, 12 C). Accuracies in percent. Global scaling entries reported at $\beta = 2.0$ for comparison.

Configuration	A	B	C
Baseline	100.0	62.5	41.7
SLB $k=25\%, \beta=2.0$	100.0	75.0	75.0
SLB $k=25\%, \beta=3.0$	83.3	50.0	83.3
SLB $k=50\%, \beta=2.0$	100.0	50.0	75.0
SLB $k=50\%, \beta=3.0$	91.7	37.5	75.0
SLB $k=75\%, \beta=2.0$	100.0	50.0	75.0
Global scaling $\beta=2.0$	100.0	50.0	75.0

D Full β Sweep Numbers

Table 15: Fine-grained β sweep on 69 C-deep and 120 novel questions at $k = 25\%$ for both backbones.

β	Gemma-2B		Mistral-7B	
	C-deep (%)	Novel (%)	C-deep (%)	Novel (%)
1.00	46.4	99.2	53.6	95.8
1.25	52.2	97.5	62.3	94.2
1.50	56.5	97.5	66.7	95.0
1.75	60.9	97.5	69.6	95.0
2.00	68.1	97.5	72.5	95.8
2.25	68.1	97.5	73.9	96.7
2.50	69.6	96.7	76.8	97.5
2.75	75.4	96.7	73.9	98.3
3.00	76.8	96.7	73.9	98.3

E Latency Measurements

Table 16 reports end-to-end generation latency on 50 KID-Bench questions timed on a single A800 80GB GPU.

Table 16: Latency of methods on 50 KID-Bench questions, single GPU, greedy decoding.

Method	Mean (ms)	Median (ms)	Std (ms)
Baseline (hypernet only)	1098.5	1056.6	133.7
SLB	1117.8	1089.6	140.0
Conflict-Aware	1759.2	1797.2	213.5

F External Validation Sets

F.1 Held-out 30 Questions

Thirty conflict questions covering countries and capitals (Germany, Canada, Italy), chemical formulas (NaCl, CO₂, Fe), planetary positions, historical dates, and companies and landmarks. No entity in this set overlaps with KID-Bench. Accuracies reported in the main text (baseline 50%, SLB 70%, CA 76.7%).

F.2 CounterFact-Style 40 Questions

Forty conflict questions in the style of CounterFact benchmark: birthplaces of historical figures, authorships of famous books, scientific discoveries, founders of tech companies, film directors, and common chemistry facts. Accuracies reported in the main text (baseline 42.5%, SLB 57.5%, CA 67.5%).

F.3 RippleEdits-Style 40 Questions

Forty conflict questions inspired by the RippleEdits benchmark format (subject, relation, target): heads of state, airport locations, scientific discoveries, chemistry symbols, famous films and composers. Accuracies: baseline 42.5%, SLB 65.0%, Conflict-Aware 77.5%. Full question list is included in the release.

G Token-Probability Probe Threshold Sweep

As a principled alternative to the lexical uncertainty markers used in CA, we replace the probe with a threshold on the base model’s maximum softmax probability at the first generated token. On 100 novel and 194 conflict questions, we obtain:

Table 17: CA token-probability probe threshold sweep on Gemma-2B. The lexical baseline (reported in main text) achieves A 97.1%, C-avg 77.4% for comparison.

Threshold	A accuracy (%)	C accuracy (%)	Boost fired on A	Boost fired on C
0.3	97.0	77.8	98 / 100	193 / 194
0.4	97.0	77.8	94 / 100	189 / 194
0.5	98.0	76.3	77 / 100	185 / 194
0.6	98.0	73.2	60 / 100	167 / 194
0.7	98.0	72.2	50 / 100	158 / 194

The optimal threshold (0.3–0.4) matches the lexical CA within 1 percentage point on both axes.

H Layer Norm Statistics

The peak-layer identity is remarkably stable across the 20 documents used for the layer-norm study. For Gemma-2B, layers 12, 13, 14, 15, 17, and 18 appear in the top tier (top 25% by $s_l = \|A_l\|_F \cdot \|B_l\|_F$) for over 80% of documents; for Mistral-7B, the corresponding top-tier layers are 7, 17, 18, 26, 28, and 30. The mean normalized s_l at the peak layer is 0.98 for Gemma and 0.74 for Mistral, with standard deviations of 0.07 and 0.11 respectively. Because the peak layers are shared across documents, a single boosting schedule (SLB at $k = 25\%$) can be applied uniformly without per-document calibration, which is what makes SLB a training-free post-processing method rather than a per-document search.

I Baseline Compatibility Barriers

We attempted three cross-editor comparisons beyond ROME on Mistral-7B (Table 9); two hit backbone/data barriers that are worth documenting so that readers interpret the comparison set correctly.

ROME on Gemma-2-2b-it. Gemma-2’s attention uses sliding-window plus logit softcapping and requires HybridCache. With `attn_implementation="eager"`, `cache_implementation=None`, and `use_cache=False` set in both `config` and `generation_config`, the model loads and runs inference, but ROME’s `rewriting_targets` scatter (`compute_v.py`, 25 gradient steps per edit) triggers a CUDA `index_copy_out-of-bounds` on all 40 RippleEdits items, stemming from a 1-token offset between the decoder-layer kv-cache position accounting and the rewriting-target index map under the eager path. We tried patching the nethook with-kwarg hook adapter (the analogous fix made ROME on Mistral-7B run cleanly, see Table 9), cache toggling at both `config` and `generate` call sites, and `bf16` versus `fp32` loads; the scatter error persisted. We document the failure as a compatibility barrier between Gemma-2’s architectural choices and ROME’s per-layer rewriting path, not as an accuracy claim.

MEMIT on Mistral-7B. MEMIT’s multi-edit update uses a covariance matrix computed over $\sim 100k$ Wikipedia tokens at the MLP input. In an offline cluster we patched `layer_stats.py` to fall back to a synthetic English corpus when the Wikipedia fetch fails; the covariance was computable but unstable, producing z -error magnitudes in the 10^6 range (compared to ~ 10 on well-calibrated covariance) and causing the edit deltas to catastrophically destroy the model: all 40 outputs degenerated into repetitions of a single Dutch token. We conclude that MEMIT requires its published Wikipedia-style covariance statistics to function; without them, the method is not comparable. This is a property of MEMIT, not of our methodology, and we report it here for reviewers considering an offline replication.

FT-per-doc across backbones. We fine-tune a LoRA adapter (rank 16, $q/k/v/o$ projections, 200 AdamW steps at $3e-4$) on each conflict document in isolation and evaluate on the paired conflict

query, using the same 20 C-deep items across three backbones. Gemma-2-2b-it reaches 82.6%, Qwen-3-4B-Instruct-2507 reaches 85.0%, and Mistral-7B-Instruct-v0.2 reaches only 70.0%: on 5 of 20 items, the per-doc LoRA overfits the short document and produces degenerate outputs (repeating “the the the”, “OOO...”, or collapsing into the document’s boilerplate). On Mistral, CA (72.5% on the full 69 C-deep, Table 1) therefore *exceeds* per-document fine-tuning by 2.5 pp on the same backbone, inverting the narrative that per-doc fine-tuning is a reliable upper bound. The pattern is backbone-dependent: on Gemma and Qwen FT-per-doc beats CA by 11–12 pp, on Mistral CA wins. FT-per-doc’s cross-backbone spread is 15 pp (70–85%) whereas CA’s is 3.7 pp (71–74.7%), so CA is the more consistent method. A 3-seed re-run on Mistral with the same per-document fine-tuning recipe further confirms FT instability: across three random seeds the accuracy is 85.5/60.9/68.1% (mean 71.5 ± 12.7 pp stdev), spanning a 24-point range from a single seed change. CA, being a single deterministic forward pass per query, has no such seed dependency. Per-edit wall clock: FT-per-doc ~ 60 s per document (per seed), CA ~ 0.5 s per query.